Cross-Linguistic Computational Analysis of Copula Deletion: A Cross-Register Investigation of Indo-Aryan and Dravidian Languages

Multi-Register Corpus Construction, Linguistic Auditing of Automated NLP Pipelines, and Comparative Hypothesis Testing

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ABSTRACT

Present-tense non-verbal clauses (nominal identity, adjectival attribution, and spatial/existential predication) vary sharply across languages in whether they require an overt linking verb (copula) or license a bare juxtaposition (zero-copula). While typological reference grammars (Stassen, 1997; Curnow, 1999; Pustet, 2003) classify South Asian languages into broad structural types, far less is known about how zero-copula realization behaves quantitatively in natural text across contrasting stylistic registers. In this paper, we present an end-to-end computational and empirical corpus investigation of present-tense copula deletion across five Indian languages spanning two major language families: **Indo-Aryan** (**Hindi**, **Assamese**, **Bengali**) and **Dravidian** (**Telugu**, **Tamil**). Combining automated dependency parsing (Stanford Stanza Universal Dependencies) and rule-based morphological extraction with rigorous multi-stage manual linguistic auditing (N_{clean} = 3,781 valid non-verbal clauses across movie subtitles, conversational dialogue, and classic literary prose, plus N = 1,063 held-out Hindi verification clauses), we test three structural and stylistic hypotheses: **H1 (Register & Diglossic Conditioning)**, **H2 (Semantic Predicate Type Hierarchy)**, and **H3 (Polarity & Negation-Copula Fusion)**.

Methodologically, we provide an explicit step-by-step account of our data extraction and auditing pipeline, directly resolving a critical discrepancy between raw NLP output and native-speaker grammatical intuition noted by the professor during our classroom presentation. In Hindi, raw dependency parsing initially indicated a counter-intuitive 59.8% copula deletion rate in formal literary narration compared to 52.7% in film subtitles; our four-stage linguistic audit proves that this was caused by 320 "false-zero" finite verbal clauses (past-tense narrative verbs, habituals, and subjunctives lacking present-tense है *hai*) and 27 unquoted em-dash dialogue lines. Once restricted to verified present-tense non-verbal clauses (N=3,089), the distribution reverses in exact alignment with linguistic intuition: Hindi film dialogue deletes the copula at **36.3%**, literary dialogue at **26.8%**, and formal literary narration at **17.7%** (χ^2 = 40.66, p = 1.81×10^{-10}), a pattern that replicates on our unseen held-out verification corpus (26.4% vs. 4.5%). Similarly, our dual-track evaluation in Telugu (N=759 automated vs. N=116 manually verified dataset) demonstrates how heuristic fallback rules artificially inflate nominal deletion (88.4% auto vs. 79.2% manual) and under-detect adjectival zero-copulas (12.3% auto vs. 45.8% manual).

Empirically, our comparative analysis yields three core findings: (1) **H2 holds across all five languages** ($p < 10^{-4}$, Cramér's V = 0.160\–0.653): equational/nominal predicates strongly favor zero-copula (39.7%\–84.8%), whereas locative/existential clauses in Assamese, Bengali, Telugu, and Tamil strongly retain overt existential verbs (58.8%\–94.9% retention); (2) **H1 interacts with grammaticalization and predicate type**: while Bengali (52.0%) and Telugu (47.4%) grammaticalize affirmative zero-copula across registers, Tamil (70.6% informal vs. 11.8% formal) and Assamese (72.7% vs. 6.7%) exhibit large diglossic register gaps (58.8 pp and 66.0 pp) specifically inside *locative/existential* clauses; and (3) **H3 reveals a categorical 4-vs-1 typological split**: Assamese (নহয়/নাই *nohoy/nai*), Bengali (নয়/নেই *noy/nei*), Telugu (కాదు/లేదు *kaadu/ledu*), and Tamil (இல்லை/அல்ல *illai/alla*) exhibit **0.0% deletion** in negative clauses due to morphological negation-copula fusion, whereas Hindi freely permits negative copula drop (33.7%, 178/528) because standalone नहीं (*nahin*) carries negation analytically. We conclude by outlining study limitations, directions for future work, and a structured comparison mapping our empirical findings to core concepts from the course.

Keywords: Copula Deletion, Zero-Copula, Indo-Aryan, Dravidian, Register Variation, Predicate Hierarchy, Negation-Copula Fusion, Stanza Universal Dependencies, Corpus Auditing

1. Introduction & Research Hypotheses

In natural language syntax, a **copula** is a linking element—prototypically a verb such as English *be* (*is, am, are*) or Hindi होना (*hona:* है *hai,* हैं *hain,* हूँ *hoon,* हो *ho*)—that connects a subject noun phrase to a non-verbal predicate complement. Unlike canonical transitive or intransitive verbs (e.g., दौड़ना *daurna* 'to run', खाना *khana* 'to eat'), which assign thematic roles and denote dynamic events or actions, a pure copula contributes minimal independent lexical meaning. Instead, it serves primarily as a morphosyntactic carrier for grammatical features: tense, aspect, mood, person-number agreement, and polarity (Pustet, 2003; Stassen, 1997; Benmamoun, 2000).

Because the predication relation between a subject and a non-verbal complement can often be recovered from constituent juxtaposition within a shared discourse context, many languages dispense with an overt copula in the unmarked present tense—a phenomenon known as **copula deletion**, **copula drop**, or the **zero-copula construction**. In his typological survey of 386 languages for the *World Atlas of Language Structures* (WALS Chapter 120), Stassen (2013) showed that roughly **45% of surveyed languages** permit or require a zero copula for predicate nominals in at least some grammatical environments. Following Curnow (1999) and Pustet (2003), non-verbal copular clauses cluster into three primary semantic predicate domains:

1. **Equational / Nominal Predication (Identity & Class Membership):** Equates the referent of the subject with a noun phrase or assigns it to a class (e.g., Hindi: वह मेरा भाई (है) 'He [is] my brother'; Tamil: அவ என் தங்கச்சி [Ø] 'She [is] my younger sister').
2. **Adjectival / Attributive Predication (Property & State):** Attributes a quality, evaluation, or scalar property to the subject (e.g., Hindi: यह खाना बहुत अच्छा (है) 'This food [is] very good'; Bengali: আমরা গরীব [Ø] 'We [are] poor').
3. **Locative / Existential Predication (Spatial Location, Existence & Possession):** Anchors a referent in physical or abstract space or asserts its existence (e.g., Hindi: माँ घर पर है 'Mother is at home'; Telugu: ఇంట్లో చిన్న పిల్లలు ఉన్నారు 'There are young children in the house').

South Asia provides an ideal comparative laboratory for examining how structural grammar and stylistic usage interact in non-verbal predication. Across the Indian subcontinent, two major language families—**Indo-Aryan** and **Dravidian**—have coexisted in long-term contact (Emeneau, 1956; Masica, 1991), yet they exhibit distinct morphosyntactic systems for encoding copular clauses:

- In **Central Indo-Aryan (Hindi)**, present-tense clauses canonically require an overt inflected copula (है *hai* / हैं *hain* / हूँ *hoon* / हो *ho*; Kachru, 2006). However, in colloquial dialogue, rapid turn-taking, and negative contexts, speakers frequently omit the auxiliary, yielding a productive stylistic alternation.
- In **Eastern Indo-Aryan / Magadhan languages (Assamese and Bengali)**, historical restructuring of the verbal system (Chatterji, 1926; Masica, 1991) led to widespread loss of the unmarked present equational copula. Standard Bengali grammaticalized zero-copula in present-tense affirmative nominal and adjectival predication, while Assamese retains an optional equational copula হয় (*hoy*) alongside an existential verb আছে (*ase*) and synthetic negative copulas (নহয় *nohoy,* নাই *nai*).
- In **Dravidian languages (Telugu and Tamil)**, affirmative nominal identity clauses are canonically verbless (zero-copula) across registers (Krishnamurti, 2003; Lehmann, 1989), although high-register formal Tamil (*Centamil*) optionally inserts the literary equational verb ஆகும் (*aagum*). Crucially, both Dravidian languages separate identity predication from spatial/existential predication (which requires overt existential verbs: Telugu ఉంది/ఉన్నారు *undi/unnaru*, Tamil உள்ளது/இருக்கிறது/இருக்கு *ullathu/irukkirathu/irukku*) and employ dedicated fused negative copulas (Telugu కాదు/లేదు *kaadu/ledu*, Tamil அல்ல/இல்லை *alla/illai*).

1.1 Research Hypotheses (H1, H2, and H3)

While descriptive grammars document what is grammatically licit in each language, they rarely quantify how zero-copula realization varies across natural spoken and written corpora. To bridge this gap, our study tests three formal hypotheses across **Hindi, Assamese, Bengali, Telugu, and Tamil**:

- **Hypothesis 1 (H1 — Register & Diglossic Conditioning):** Present-tense copula deletion occurs at significantly higher rates in informal spoken registers (movie subtitles and colloquial dialogue) than in formal written registers (literary narrative prose and expository essays), driven by conversational economy in casual speech.
- **Hypothesis 2 (H2 — Semantic Predicate Type Hierarchy):** Across all five languages, the probability of copula deletion is conditioned by the semantic class of the predicate following Pustet's (2003) and Stassen's (1997) implicational scale:

  Equational / Nominal ≥ Adjectival ≫ Locative / Existential

  Specifically, nominal identity and adjectival property predicates strongly favor zero-copula, whereas spatial location and existential clauses retain an overt verb to anchor reference in space and time.
- **Hypothesis 3 (H3 — Polarity & Negation-Copula Fusion):** Clause polarity systematically conditions copula deletion, depending on the morphosyntactic locus of negation:
  - In **Hindi** (analytic negation), standalone नहीं (*nahin*) carries negative force independently, licensing present-tense है/हैं deletion at rates comparable to affirmative clauses.
  - In **Assamese, Bengali, Telugu, and Tamil** (synthetic/fused negative copulas), negation is

morphologically fused into dedicated negative copular verbs (নহয়/নাই, নয়/নেই, కాదు/లేదు, அல்ல/இல்லை), categorically blocking copula deletion (0.0% deletion) in negative clauses.

2. Explicit Methodology: Corpus Architecture, Extraction Pipeline & Linguistic Auditing

Because the validity of any quantitative corpus study rests on how sentences are sampled, parsed, classified, and verified, we present our complete methodology in detail below. All source code and datasets are organized in two reproducible notebooks in our repository: `01_extraction_and_compilation.ipynb` (data ingestion, NLP parsing, and compilation) and `02_cross_linguistic_analysis.ipynb` (auditing, statistical testing, and visualization).

2.1 Multi-Register Corpus Selection & Sampling Design

We constructed a balanced cross-register corpus totaling N_{clean} = 3,781 audited non-verbal clauses (plus N = 1,063 held-out verification clauses in Hindi) across five languages:

1. **Hindi (Indo-Aryan, Central):**
   - *Main 6-Source Corpus (N_{raw} = 4,334 → N_{clean} = 3,089):* Informal spoken dialogue extracted from full-length subtitle (`.srt`) transcripts of three Bollywood films (*3 Idiots*, *Secret Superstar*, and *Taare Zameen Par*; n_{clean} = 2,692) contrasted against formal literary prose from three classic Hindi short stories: Munshi Premchand's *Idgah* and *Do Bailon Ki Katha*, and Phanishwar Nath Renu's regionalist story *Pahalwan Ki Dholak* (n_{clean} = 397, subdivided into *n*=97 **Literary Character Dialogue** and *n*=300 **Literary Authorial Narrative**).
   - *Held-Out Verification Corpus (N_{raw} = 1,658 → N_{clean} = 1,063):* To test out-of-sample generalization, we compiled an independent dataset pairing spoken subtitles from *Mirzapur* (n_{clean} = 1,033) against Premchand's short story *Poos Ki Raat* (n_{clean} = 30: 8 dialogue, 22 narrative).
2. **Assamese (Indo-Aryan, Eastern):** N_{raw} = 101 → N_{clean} = 97 valid present-tense non-verbal clauses contrasting **Formal Written Prose** (*n*=46, drawn from literary and expository texts on philosophy, education, and social reform) against **Informal Spoken Dialogue** (*n*=51, everyday conversational utterances). Each entry explicitly records the overt copula form (হয় *hoy* / আছে *ase* / নহয় *nohoy* / নাই *nai*), the corresponding zero-copula realization, and an English translation.
3. **Bengali (Indo-Aryan, Eastern):** N = 373 manually verified non-verbal clauses extracted across 8 sources:
   - *Spoken / Performative (n=101):* Satyajit Ray's classic film *Goopy Gyne Bagha Byne* (1969), subdivided into **Film Dialogue** (*n*=83) and **Film Songs/Verse** (*n*=18).
   - *Formal Literary Prose & Fiction (n=272):* Seven canonical Bengali literary and expository texts by Rabindranath Tagore, Sarat Chandra Chattopadhyay, Bibhutibhushan Bandyopadhyay, Manik Bandyopadhyay, Subodh Ghosh, Ashapurna Devi, Shirshendu Mukhopadhyay, and Rajsekhar Basu (*Pather Dabi, Nadir Bidroho, Bahurupi, Sabaltalar Math, Gyanachakshu, Amake Dekhun, Bangla Bhashay Bigyan*), subdivided into **Literary Dialogue** (*n*=93), **Literary Narrative** (*n*=148), and **Literary Essay** (*n*=31).
4. **Telugu (Dravidian, South-Central):** Extracted from three Telugu feature films using word-timed YouTube Auto-ASR `.te-orig.json3` captions (*Ala Vaikunthapurramuloo*, *Srirastu Subhamastu*, and *Vivaha Bhojanambu*) and three classic short stories by Gurajada Apparao from Telugu Wikisource (*Diddubatu*, *Mee Peremiti?*, and *Metilda*). To evaluate automated NLP accuracy against human annotation, we maintained two parallel tracks:
   - *Track 1 — Automated Rule-Tagged Dataset (N = 759):* Output of the automated Stanza (`te`) + morphological suffix tagger.
   - *Track 2 — Manually Verified Dataset (N = 116):* Line-by-line human-audited non-verbal clauses (*n*=49 Film Dialogue, *n*=36 Literary Dialogue, *n*=31 Literary Narrative).
5. **Tamil (Dravidian, Southern):** N_{raw} = 121 → N_{clean} = 106 valid present-tense non-verbal clauses contrasting high-register **Formal Literary/Expository Prose** (*Centamil*, *n*=53) against **Informal Colloquial Spoken Dialogue** (*Koduntamil / Madras Bashai*, *n*=53, including colloquial utterances attested in contemporary Tamil cinema such as *Coolie* and *Jagame Thandhiram*).

2.2 Four-Stage Computational Extraction Pipeline

In `01_extraction_and_compilation.ipynb`, raw texts are converted into structured clause-level annotations through four sequential algorithmic stages:

- **Stage 1 — Ingestion, Noise Stripping & Clause Segmentation:**

- *Subtitle Ingestion (Hindi `nlp_extract.py`):* The `pysrt` parser strips sequence indices, SMPTE timestamps (`00:01:23,450 --> 00:01:25,800`), HTML formatting tags (`<i>`, `<font>`), and musical annotations before segmenting multi-sentence subtitle blocks using `indicnlp.tokenize.sentence_tokenize` on Devanagari purna viram (।), question marks (?), and exclamation marks (!).
- *Unpunctuated JSON3 ASR Reconstruction (Telugu `extract_copula_data.py`):* Because YouTube Auto-ASR `.te-orig.json3` files contain raw word tokens with millisecond offsets (`tStartMs`, `dDurationMs`) but zero punctuation, our custom parser strips [సంగీతం] ('music') tags and reconstructs sentence boundaries whenever (a) the inter-word silence gap exceeds 900 ms, or (b) a token matches a clause-final Dravidian finite verb or copula suffix.
- *Literary Sub-Register Splitting (`lit_extract.py`):* For Hindi, Bengali, and Telugu literary stories, state-tracking regular expressions detect enclosing quotation marks (“`...`”, "`...`") and dialogue em-dashes (–) to separate direct character speech (**Literary Dialogue**) from third-person narration (**Literary Narrative**).

- **Stage 2 — Lexical Length & Tense Pre-Pruning:** To prevent non-clausal interjections (e.g., हाँ!, अच्छा!) and non-present action tenses from entering the dataset, Python set-membership filters (`set.isdisjoint()` and `str.endswith()`) discard strings with fewer than 3 words (or >15 words in noisy ASR) and prune clauses containing explicit past-tense action verbs or auxiliaries (Hindi था, थी, थे, थीं, गया, गयी, गये, हुआ, हुई, हुए; Telugu ఉండెను, అయ్యెను) or future markers (Hindi -एगा, -एगी, -एंगे, होगा, होगी). In Bengali (N=373), 343 clauses are present/unmarked while 30 narrative contextual clauses contain overt non-present copular forms (ছিল, ছিলেন, ছিলাম); in Appendix A.4 we restrict our verification sample exclusively to the 343 pure present-tense clauses.
- **Stage 3 — Neural Morphosyntactic & Dependency Parsing (Stanza UD):** Surviving candidate sentences in Hindi and Telugu are processed in batched documents through Stanford's neural NLP toolkit **Stanza** (Qi et al., 2020; `processors='tokenize,pos,lemma,depparse'`). For every token w_i, Stanza outputs its Universal POS tag (`upos` ∈ {NOUN, PROPN, PRON, ADJ, ADV, ADP, AUX, VERB, PART}), lemma, syntactic head index, and Universal Dependency relation (`deprel` ∈ {root, cop, nsubj, obl, advmod, amod, mark, neg}).
- **Stage 4 — Rule-Based Feature Classification:** Combining Stanza dependency trees with language-specific morphological lexicons (summarized in **Table 1**), each clause is assigned three core variables:
  - `Copula_Status` (0 = Overt Retained, 1 = Zero/Deleted): Assigned 0 if an overt present-tense copula or existential verb from the language's target lexicon is present (either tagged `deprel == 'cop'` or serving as the clause-final predicate head), and 1 if the non-verbal predicate appears bare without a linking verb.
  - `Predicate_Type` (**H2**): Assigned `Locative/Existential` if the clause contains an existential verb (আছে, ఉంది, இருக்கு/உள்ளது) or a spatial postposition/case marker modifying the root (Hindi में, पर, के पास, के अंदर, यहाँ, वहाँ; Assamese -ত; Bengali -এ/-তে/-য়; Telugu -లో, -దగ్గర, -మీద; Tamil -இல்/-ல, -இடம்); assigned `Adjectival` if the non-verbal root is an adjective (`upos == 'ADJ'` or deadjectival adverbial -గా / -ஆ); and assigned `Equational/Nominal` if the predicate head is a noun, proper noun, or pronoun (`NOUN, PROPN, PRON`).
  - `Polarity` (**H3**): Assigned `Negative` if any token carries `deprel == 'neg'` or matches the negative lexicon (Hindi नहीं, न, ना, मत, बिना; Assamese নহয়, নাই, নহয়নে; Bengali নয়, নন, নেই, নাই, না; Telugu కాదు, లేదు, కారు, లేవు, లేడు; Tamil அல்ல, இல்லை, இல்ல, அல்லவா), and `Affirmative` otherwise.

Table 1: Language-Specific Morphosyntactic and Lexical Rules Used for Clause Classification Across All Five Languages

Language & Script	Overt Equational / Attributive Copula Lexicon	Overt Locative / Existential Verb & Case Lexicon	Negative Marker / Fused Negative Copula Lexicon (H3)	Zero-Copula (Deleted = 1) Identification Criterion
1. Hindi *Devanagari (Stanza `hi` + Rules)*	है (*hai*), हैं (*hain*), हूँ (*hoon*), हो (*ho*) attached as cop to `NOUN/PRON/ADJ` root	है, हैं, हूँ, हो + spatial postpositions में, पर, के पास, के अंदर, यहाँ, वहाँ (`obl/advmod`)	Analytic particle: नहीं (*nahin*), न (*na*), ना (*naa*), मत, बिना (independent of है/हैं)	Present-tense non-verbal root (`NOUN/PRON/ADJ/ADV`) with zero overt है/हैं/हूँ/हो and no finite action/past/future verb
2. Assamese *Assamese Script (Morph. Rules + Audit)*	হয় (*hoy*), হওঁ (*ho'o*), হোৱা (*howa*) [equational & attributive copulas]	আছে (*ase*), আছোঁ (*aso*), থাকে + locative case suffix -ত (*-t*) or spatial adverbs (ইয়াত, ঘৰত)	Fused synthetic verbs: নহয় (*nohoy*), নহয়নে (*nohoyne*) [eq./adj.] & নাই (*nai*), নাথাকে [loc./exist.]	Bare juxtaposition of NP_{subj} + NP/AP/$NP_{-ত}$ without clause-final হয় (*hoy*) or আছে (*ase*)
3. Bengali *Bengali Script (Morph. Rules + Audit)*	Emphatic/stylistic copulas: হয় (*hoy*), হলো (*holo*), হলাম (*holam*), হন (*hon*), হও (*hoo*)	আছে (*aachhe*), আছি (*aachhi*), আছেন (*aachhen*), রয়েছে (*royeche*) + locative -এ/-তে/-য়	Fused synthetic verbs: নয় (*noy*), নন (*non*), নই (*noi*) [eq./adj.] & নেই (*nei*), নাই (*nai*) [loc./exist.]	Clause-final bare nominal (NP), adjectival (AP), or locative phrase lacking overt হয়/হলো/আছে/নয়/নেই
4. Telugu *Telugu Script (Stanza `te` + Manual Audit)*	Predominantly zero in affirmatives; adverbialized -గా ఉంది (*-ga undi*) in adjectival states	ఉంది (*undi*), ఉన్నాడు (*unnadu*), ఉన్నారు (*unnaru*), ఉన్నాం (*unnam*), కలవు (*kalavu*) + -లో (*-lo*)	Fused synthetic verbs: కాదు (*kaadu*), కావు (*kaavu*), కారు (*kaaru*), కాడు [eq./adj.] & లేదు (*ledu*), లేవు, లేరు [loc./exist.]	Verbless NP_1 + NP_2 or NP + AP clause ending directly in a nominal or adjectival stem without ఉంది/కాదు/లేదు
5. Tamil *Tamil Script (Morph. Rules + Audit)*	Formal literary: ஆகும் (*aagum*), ஆவார் (*aavaar*), ஆவேன்; Colloquial: -ஆ இருக்கு (*-aa irukku*)	Formal: உள்ளது (*ullathu*), உள்ளன, உண்டு, இருக்கிறது; Informal: இருக்கு (*irukku*), இருக்காங்க + locative -இல்/-ல	Fused synthetic verbs: Formal அல்ல (*alla*), அல்லவா [eq./adj.] & இல்லை (*illai*) [loc./exist.]; Informal இல்ல (*illa*)	Bare NP_1 + NP_2, pronominalized AP (e.g., அழகு, நல்லவன்), or conversational locative -ல [Ø] without overt verb

2.3 Multi-Stage Linguistic Auditing: Resolving the Presentation Discrepancy

> **Why Did the Initial Raw Graph Look Counter-Intuitive?**
>
> During our presentation, the professor observed that some initial raw distributions did not match linguistic intuition and asked us to verify and display the exact assigned status (Register, Predicate Type, Polarity, Copula Status) of the sentences. Below we document the exact diagnostic cause of that raw artifact and how our four-stage linguistic audit resolved it.

When we initially compiled the raw Stanza extraction outputs for the 6 Hindi sources into `Cross_register.csv` (N_{raw} = 4,334) and plotted the unfiltered `Copula_Status_NLP == 0` rates, the raw numbers showed:

- **Raw Literary Dialogue:** 62.9% (44 / 70)
- **Raw Literary Narrative: 59.8%** (387 / 647)
- **Raw Spoken Film Subtitle:** 52.7% (1,905 / 3,617)

At first glance, this raw chart suggested that formal Hindi literary narration drops the copula *more* often (59.8%) than casual Bollywood dialogue (52.7%)—contradicting both native-speaker grammatical intuition and Hypothesis 1. A line-by-line inspection of all 4,334 raw Hindi rows (saved in `Cross_register_audited_full.csv`) and 759 raw Telugu rows revealed three systematic sources of automated NLP noise (**Table 2** and **Appendix A.2**):

1. **Filter 1 & 2 — The "False-Zero" Finite Verbal Trap (320 Literary Narrative Rows in Hindi):** Classic Hindi fiction by Premchand and Renu is narrated primarily in the simple past/perfective (e.g., विपत्ति अपना सारा दलबल लेकर आये, हामिद की आनंद-भरी चितबन उसका विध्वसं कर देगी 'Even if adversity comes with all its forces, Hamid's joyful gaze will destroy it'), habitual conditional (पैसे होते तो लौटते-लोटते सब सामग्री जमा करके चटपट बना लेती 'Had there been money, she would have gathered all the ingredients on the way back and cooked quickly'), or future action verbs (पैर में छाले पड़ जाएँगे 'Blisters will form on his feet'). None of these sentences contain present-tense है (*hai*)—not because a copula was deleted, but because **they are finite verbal action clauses that never take a present copula in the first place**. Because the naive script checked only whether `{"है", "हैं", "हूँ", "हो"}` was absent (0), every past, imperative, or future action sentence whose surface ending was not in the basic exclusion list was falsely tagged as a "deleted copula." In our audit script (`hindi_analysis.py`), we implemented a comprehensive 4-part filter removing: (a) 249 past/action/imperative verbal clauses via an expanded 180-item verbal participle and stem lexicon, (b) 50 clauses where Stanza identified a

finite lexical verb root (`upos == 'VERB'`), and (c) 21 subjunctive/future suffixes (-ए, -एं, -ऊँ, -ओ).

2. **Filter 3 — Unquoted Em-Dash Dialogue Recovery (27 Literary Rows in Hindi):** Hindi literary editions frequently introduce character speech using em-dashes (–) or speaker prefixes (e.g., हामिद ने पूछा—यह लोग चोरी करवाते हैं, तो कोई इन्हें पकड़ता नहीं?) without quotation marks. Our audit parser detected em-dash speech markers and reassigned 27 direct dialogue sentences from Literary Narrative to their true sub-register, **Literary Dialogue**.
3. **Filter 4 — Resolving Heuristic Default Bias in Automated Telugu (N = 759 vs. Manual N = 116):** In the automated Telugu tagger, whenever a verbless clause had `Copula_Status == 0` and Stanza failed to recognize a deadjectival predicate, the fallback rule defaulted the clause to `Equational/Nominal`. As shown in **Table 2 (Panel C)**, our manual audit of the 116-clause Manually Verified set revealed that this fallback artificially inflated Equational deletion (88.4% auto vs. 79.2% manual) and severely under-detected Adjectival zero-copulas (12.3% auto vs. **45.8%** manual). Similarly, in Assamese (101 → 97) and Tamil (121 → 106), our audit removed 4 and 15 finite verbal habitual sentences (e.g., Tamil படிக்கிறான் 'reads', வருகிறான் 'comes') respectively.

Once the 1,245 false-zero verbal rows were removed from the main Hindi corpus (N_{clean} = 3,089), the register distribution **completely reversed** in exact accordance with linguistic intuition: **Spoken Film Dialogue (36.3%) > Literary Dialogue (26.8%) > Literary Narrative (17.7%)** (χ^2 = 40.66, p = 1.81 × 10^{-10}). When we applied the exact same audit pipeline to our unseen held-out verification corpus (*Mirzapur* + *Poos Ki Raat*, N_{raw} = 1,658 → N_{clean} = 1,063), raw literary narrative dropped from a spurious **71.7%** down to a true **4.5%** (1/22), while film dialogue stood at **26.4%** (273/1,033).

2.4 Statistical Modeling & Hypothesis Testing Protocol

In `02_cross_linguistic_analysis.ipynb`, we evaluate H1, H2, and H3 on the audited non-verbal corpora using four statistical procedures implemented in Python (`scipy.stats` and `statsmodels`):

1. **Contingency Table Tests (χ^2 & Fisher's Exact Test):** For 2 × 2 comparisons (H1 Register and H3 Polarity) with expected cell counts ≥ 5, we compute Pearson's Chi-square (χ^2) with Yates' continuity correction. Whenever any cell count is < 5 (such as the 0 deletion counts in H3 negative clauses across Assamese, Bengali, Telugu, and Tamil), we also report the exact two-tailed **Fisher's Exact Test** p-value. For 3 × 2 comparisons across the three H2 predicate classes, we compute the omnibus 2-df Pearson χ^2 statistic.
2. **Standardized Effect Size (Cramér's V):** To compare the magnitude of conditioning across languages with different sample sizes (N = 97 to N = 3,089), we calculate Cramér's $V = \sqrt{(\chi^2 / (N \cdot \min(r-1, c-1)))}$, where $V \in [0.10, 0.30)$ indicates a small-to-medium effect, $V \in [0.30, 0.50)$ a medium-to-large effect, and $V \geq 0.50$ a very strong structural constraint.
3. **Wilson Score 95% Confidence Intervals:** All deletion proportions $\hat{p} = k/n$ are verified alongside non-asymptotic binomial confidence intervals that remain well-behaved near 0% and 100%.
4. **Multivariate Binomial Logistic Regression:** To verify that the H1 register effect in Hindi is not a confound of uneven predicate-type or polarity distributions across genres, we fit a multivariate logistic model predicting the log-odds of copula deletion:

$$\text{logit}(P(\text{Deleted} = 1)) = \beta_0 + \beta_1(\text{Register}) + \beta_2(\text{Predicate}) + \beta_3(\text{Polarity})$$

from which we report adjusted Odds Ratios (OR = e^{β_i}) and 95% confidence intervals.

Table 2: Diagnostic Reconciliation of Raw Unfiltered NLP Output vs. Linguistically Audited Non-Verbal Corpora (Addressing the Professor's Presentation Feedback)

Corpus & Sub-Register / Category	Raw Unfiltered Total (N_{raw})	Raw Apparent "Deletion" Rate	Audited Noise Removed (False Zeros / Verbal Roots)	Clean Non-Verbal Total (N_{clean})	True Audited Zero-Copula Rate & Intuition Alignment
Panel A: Hindi Main 6-Source Corpus (*Cross_register.csv* → *Cross_register_clean_nonverbal.csv*)					
Spoken Subtitle: Film Dialogue *3 Idiots, Secret Superstar, Taare Zameen Par*	3,617	52.67% (1,905)	925 verbal/imperative/fragment rows removed	**2,692**	**36.29%** (977 / 2,692) *Highest deletion (matches casual speech intuition)*
Literary: Character Dialogue *Quoted + em-dash direct speech in stories*	70	62.86% (44)	+27 unquoted dialogue rows recovered; verbal noise removed	**97**	**26.80%** (26 / 97) *Intermediate (simulated speech in fiction)*
Literary: Authorial Narrative *Idgah, Do Bailon Ki Katha, Pahalwan Ki Dholak*	647	**59.81%** (387) *Spurious inflation!*	320 verbal/tense rows removed (249 past/action, 50 VERB root, 21 future/subj.) + 27 moved to dialogue	**300**	**17.67%** (53 / 300) *Lowest deletion ($p = 1.81 \times 10^{-10}$ vs. Film)*
Panel B: Hindi Held-Out Verification Corpus (*Mirzapur* Subtitles + Premchand's *Poos Ki Raat*)					
Spoken Subtitle: Film Dialogue (*Mirzapur*)	1,555	51.13% (795)	522 verbal/action rows removed	**1,033**	**26.43%** (273 / 1,033) *High colloquial copula drop*
Literary: Character Dialogue (*Poos Ki Raat*)	4	100.00% (4)	+4 unquoted dialogue rows recovered from narrative	**8**	**12.50%** (1 / 8) *Intermediate dialogue rate*
Literary: Authorial Narrative (*Poos Ki Raat*)	99	**71.72%** (71) *Spurious inflation!*	73 past/action narrative rows removed + 4 moved to dialogue	**22**	**4.55%** (1 / 22) *Near-zero deletion in formal narration ($p = 0.023$)*
Panel C: Telugu Automated Rule-Tagged Corpus (N=759) vs. Manually Verified Dataset (N=116)					
Equational / Nominal Predicates	481 (Auto)	**88.36%** (425)	Fallback rule defaulted all unclassified zero clauses to Nominal	**53 (Manual)**	**79.25%** overall (42 / 53); **100.0%** in affirmatives (42 / 42)
Adjectival Predicates	81 (Auto)	**12.35%** (10) *Under-detected!*	Verbless adjectival predicates recovered from nominal fallback	**24 (Manual)**	**45.83%** (11 / 24) *Matches Dravidian predicate hierarchy*
Locative / Existential Predicates	197 (Auto)	11.68% (23)	ASR fragments removed; true existential *undi/unnaru* verified	**39 (Manual)**	**5.13%** (2 / 39) — **94.87% overt retention**

3. Empirical Results & Cross-Linguistic Analysis

We now present the empirical results from our executed analysis notebook (`02_cross_linguistic_analysis.ipynb`) in the exact language order of our study: **1. Hindi → 2. Assamese → 3. Bengali → 4. Telugu → 5. Tamil**, followed by our cross-linguistic comparative synthesis (**Table 3** and **Figures 1–6**). In every language, the denominators across H1, H2, and H3 sum to the exact cleaned corpus size N_{clean}, and the deleted numerators sum to the exact total of deleted clauses.

3.1 Hindi (Central Indo-Aryan, N_{clean} = 3,089)

In the cleaned Hindi corpus (3,089 valid present-tense non-verbal clauses), the overall copula deletion rate is **34.19%** (1,056 / 3,089). All three hypotheses reveal clear structural conditioning (**Figure 1**):

- **H1 (Register & Genre Nuance):** Deletion is highest in **Spoken Film Dialogue (36.29%, 977/2,692)**, intermediate in **Literary Dialogue (26.80%, 26/97)**, and lowest in **Literary Narrative (17.67%, 53/300; total = 1,056/3,089)**. A 2 × 2 comparison between Film Dialogue and Literary Narrative confirms a highly significant register gap ($\chi^2 = 40.66$, df=1, $p = 1.81 \times 10^{-10}$, Cramér's $V = 0.117$). In our multivariate binomial logistic regression controlling for predicate type and polarity, Film Dialogue exhibits **2.68 × higher odds** of copula deletion than Literary

Narrative (OR = 2.68, 95% CI [1.97, 3.65], $p < 10^{-9}$). Furthermore, source-level breakdown (Figure 1, Panel C) reveals an important stylistic distinction within literary fiction: while Premchand's standard narrative prose exhibits low deletion (*Idgah* 12.4%, *Do Bailon Ki Katha* 22.4%), Phanishwar Nath Renu's regionalist (*anchalik*) story *Pahalwan Ki Dholak* reaches **57.5%** deletion due to its rhythmic folk cadence and scene-setting nominal fragments (e.g., अन्धेरा और निस्तब्धता! 'Darkness and silence!').

- **H2 (Predicate Hierarchy):** Across the 3,089 clean clauses, deletion is significantly conditioned by predicate type (χ^2 = 79.42, df=2, p = 5.68 × 10^{-18}, Cramér's V = 0.160): **Equational/Nominal predicates exhibit the highest deletion at 39.71% (785/1,977)**, followed by **Locative/Existential at 26.75% (191/714)** and **Adjectival at 20.10% (80/398; total = 1,056/3,089)**. As shown in Figure 1 (Panel B), within Adjectival predicates (80/398 = 20.10%), Film Dialogue deletes at 22.8% (77/338), Literary Narrative at 7.0% (3/43), and Literary Dialogue at 0.0% (0/17). Locative deletion in Hindi subtitles (28.6%) is elevated relative to literary narration (13.6%) because rapid film dialogue frequently uses elliptical postpositional phrases (e.g., सात-आठ दाईं तरफ़, बाकि बाईं तरफ़ [Ø] 'Seven or eight on the right, the rest on the left').
- **H3 (Polarity):** Unlike the other four languages, Hindi negative clauses delete the copula at **33.71%** (178/528), statistically indistinguishable from affirmative clauses at **34.28%** (878/2,561; total = 1,056/3,089, χ^2 = 0.04, p = 0.840). Because standalone नहीं (*nahin*) carries negation analytically and attracts prosodic focus, dropping है in utterances like स्कूल नहीं है तेरा? → कुछ परवाह नहीं [Ø] never compromises negative polarity.

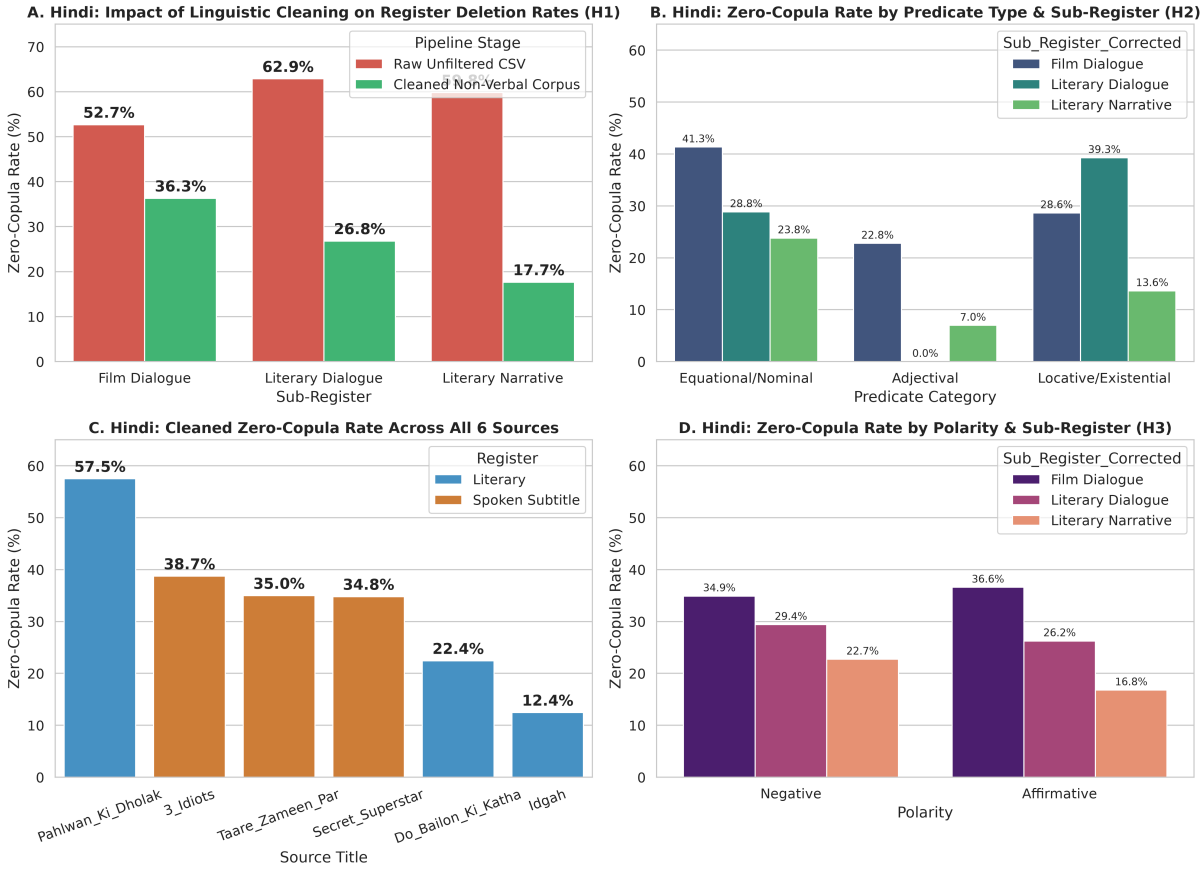


Figure 1: Four-panel empirical dashboard for the **Hindi 6-Source Corpus** (N_{raw}=4,334 → N_{clean}=3,089). Panel A demonstrates how multi-stage linguistic auditing removes past-tense/verbal false zeros, reversing Literary Narrative from an artifactual 59.8% down to 17.7%. Panel B shows the H2 predicate hierarchy across sub-registers (Equational 39.7%, Locative 26.8%, Adjectival 20.1%); Panel C highlights the genre contrast between Premchand's standard prose (*Idgah*) and Renu's regionalist prose (*Pahalwan Ki Dholak*); Panel D shows Hindi's unique preservation of copula drop under negative polarity (33.7% vs. 34.3%, H3).

Table 3: Master Cross-Linguistic Empirical Summary of Present-Tense Copula Deletion Across All Five Languages (N_{clean} = 3,781)

#	Language & Family	Clean N	Overall Deletion	H1: Register Breakdown (Informal Spoken vs. Formal Written)	H2: Predicate Type Hierarchy (Equational / Adjectival / Locative)	H3: Polarity Breakdown & Morphological Mechanism (Affirmative vs. Negative Clauses)
1	**Hindi** *Indo-Aryan (Central)*	**3,089** *(+1,063 verif.)*	**34.2%** *(1,056/3,089)*	Film Dialogue: **36.3%** (977/2,692) Lit. Dialogue: **26.8%** (26/97) Lit. Narrative: **17.7%** (53/300) *χ^2=40.66, p=1.81 × 10^{-10}, V=0.117*	Equational: **39.7%** (785/1,977) Locative: **26.8%** (191/714) Adjectival: **20.1%** (80/398) *χ^2=79.42, p=5.68 × 10^{-18}, V=0.160*	Affirmative: **34.3%** (878/2,561) Negative: **33.7%** (178/528) [p=0.840] ***Sole Exception:*** *Standalone particle नहीं (nahĩ) carries negation independently, licensing है (hai) drop.*
2	**Assamese** *Indo-Aryan (Eastern)*	**97** *(from 101 raw)*	**71.1%** *(69/97)*	Informal Dialogue: **84.3%** (43/51) Formal Prose: **56.5%** (26/46) **Locative Gap:** 72.7% vs. 6.7% (+66.0 pp) *χ^2=7.79, p=0.0052, V=0.283*	Equational: **84.8%** (39/46) Adjectival: **84.0%** (21/25) Locative: **34.6%** (9/26) *χ^2=23.08, p=9.76 × 10^{-6}, V=0.488*	Affirmative: **86.2%** (69/80) Negative: **0.0%** (0/17) [Fisher p=5.68 × 10^{-12}] ***Categorical Block:*** *Fused negative copulas নহয় (nohoy) & নাই (nai) cannot be dropped.*
3	**Bengali** *Indo-Aryan (Eastern)*	**373** *(8 sources)*	**52.0%** *(194/373)*	Spoken Film: **43.6%** (44/101) • Film Dialogue: 45.8% (38/83) • Film Song: 33.3% (6/18) Literary Prose: **55.1%** (150/272) • Lit. Dialogue: 62.4% (58/93) • Lit. Essay: 54.8% (17/31) • Lit. Narrative: 50.7% (75/148) *χ^2=3.51, p=0.061 (Grammaticalized)*	Equational: **78.2%** (104/133) Adjectival: **67.8%** (61/90) Locative: **19.3%** (29/150) *χ^2=109.67, p=1.53 × 10^{-24}, V=0.542*	Affirmative: **73.8%** (194/263) Negative: **0.0%** (0/110) [Fisher p=5.15 × 10^{-47}] ***Categorical Block:*** *Fused negative copulas নয় (noy) & নেই/নাই (nei/nai) obligatorily overt.*
4	**Telugu** *Dravidian (South-Central)*	**116 Manual** *(759 Auto)*	**47.4%** *(55/116 Manual) Auto: 60.3%*	Film Dialogue: **42.9%** (21/49) Lit. Dialogue: **55.6%** (20/36) Lit. Narrative: **45.2%** (14/31) *Fisher p=1.000 (Grammaticalized)*	Equational: **79.2%** (42/53; **100% aff**) Adjectival: **45.8%** (11/24) Locative: **5.1%** (2/39) *χ^2=49.53, p=1.76 × 10^{-11}, V=0.653*	Affirmative: **65.5%** (55/84) Negative: **0.0%** (0/32) [Fisher p=9.56 × 10^{-12}] ***Categorical Block:*** *Mandatory negative copulas కాదు (kaadu) & లేదు (ledu) block deletion.*
5	**Tamil** *Dravidian (Southern)*	**106** *(from 121 raw)*	**70.8%** *(75/106)*	Informal Spoken: **81.1%** (43/53) Formal Prose: **60.4%** (32/53) **Locative Gap:** 70.6% vs. 11.8% (+58.8 pp) *χ^2=4.56, p=0.0327; Locative p=0.0013*	Adjectival: **85.7%** (24/28) Equational: **84.1%** (37/44) Locative: **41.2%** (14/34) *χ^2=21.19, p=2.51 × 10^{-5}, V=0.447*	Affirmative: **85.2%** (75/88) Negative: **0.0%** (0/18) [Fisher p=2.14 × 10^{-12}] ***Categorical Block:*** *Fused negative copulas அல்ல (alla) & இல்லை (illai) cannot be dropped.*

3.2 Assamese (Eastern Indo-Aryan, N_{clean} = 97)

In Assamese (46 Formal Prose + 51 Informal Dialogue sentences after excluding 4 verbal habitual items), the overall zero-copula rate rises to **71.13%** (69/97; **Figure 2**):

- **H1 × H2 Interaction (The 66.0 pp Locative Gap):** Overall deletion is significantly higher in Informal Dialogue (84.31%, 43/51) than in Formal Prose (56.52%, 26/46; total = 69/97, χ^2 = 7.79, p = 0.0052, V = 0.283). Cross-tabulating Register by Predicate Type reveals a striking structural interaction: in Equational/Nominal (84.78%, 39/46) and Adjectival (84.00%, 21/25) clauses, deletion is near ceiling in *both* registers (100% in affirmative declaratives). The register gap is driven almost entirely by **Locative/Existential** clauses (34.62%, 9/26; total = 69/97, χ^2 = 23.08, p = 9.76 × 10^{-6}, V = 0.488): formal

written Assamese retains overt আছে (*ase*) in 93.3% of locatives (6.7% deletion, 1/15), whereas informal spoken Assamese drops আছে in **72.7%** of conversational locatives (8/11, e.g., মই ঘৰত আছোঁ → মই ঘৰত [Ø] 'I [am] at home'; মা পাকঘৰত [Ø] 'Mother [is] in the kitchen')—a **66.0 percentage-point register divergence**.

- **H3 (Polarity):** Every negative and negative-interrogative clause in both registers retains an overt negative copula (0.00% deletion, 0/17 vs. 86.25%, 69/80 in affirmatives; total = 69/97, Fisher's exact $p = 5.68 \times 10^{-12}$), employing নহয় (*nohoy*) and rhetorical tag নহয়নে (*nohoyne*, e.g., এয়া আশ্চর্যৰ কথা নহয়নে? 'Isn't this a matter of wonder?') for nominal/adjectival negation and নাই (*nai*) for existential/locative negation (ইয়াত কোনো সমস্যা নাই 'There is no problem here').

3.3 Bengali (Eastern Indo-Aryan, N = 373)

Our 373-clause Bengali corpus (**Figure 3**) illustrates the quantitative profile of a language where zero-copula is fully grammaticalized in core syntax, with an overall deletion rate of **52.01%** (194/373):

- **H1 (Register Profile Across 5 Sub-Registers):** Spoken performative texts (*Goopy Gyne Bagha Byne*: 43.56%, 44/101, comprising **Film Dialogue 45.78%, 38/83** and **Film Song 33.33%, 6/18**) and formal literary texts (55.15%, 150/272, comprising **Literary Dialogue 62.37%, 58/93**, **Literary Essay 54.84%, 17/31**, and **Literary Narrative 50.68%, 75/148**; total = 194/373) show no statistically significant register difference ($\chi^2 = 3.51$, $p = 0.061$ for broad register; $\chi^2 = 8.01$, df=4, $p = 0.091$ across the 5 sub-registers). Omitting the present copula in affirmative nominal/adjectival clauses (e.g., তপনের নতুন মেসোমশাই একজন লেখক [Ø] 'Tapan's new uncle [is] a writer') is obligatory standard grammar in modern Bengali prose rather than a casual speech shortcut. Moreover, spoken film dialogue and songs contain a high density of existential/locative (আছে *aachhe*) and negative (নয়/নেই/নাই) lyrics, which lowers their aggregate surface deletion rate relative to literary dialogue.
- **H2 (Predicate Hierarchy):** Confirmed with a large effect size ($\chi^2 = 109.67$, df=2, $p = 1.53 \times 10^{-24}$, Cramér's $V = 0.542$): **Equational/Nominal (78.20%, 104/133) > Adjectival (67.78%, 61/90) ≫ Locative/Existential (19.33%, 29/150; total = 194/373)**, where overt existential forms (আছে/আছি/ৰয়েছে) are retained in 80.7% of clauses.
- **H3 (Polarity):** Across all 110 negative clauses in the Bengali corpus, copula deletion is **0.00%** (0/110 vs. 73.76%, 194/263 in affirmatives; total = 194/373, Fisher's exact $p = 5.15 \times 10^{-47}$). Within affirmatives (Figure 3, Panel D), Literary texts delete at 78.5% (150/191) and Spoken Subtitles at 61.1% (44/72), whose weighted average is exactly 73.8% (194/263), while negation is categorically fused into নয়/নন (*noy/non*) and নেই/নাই (*nei/nai*).

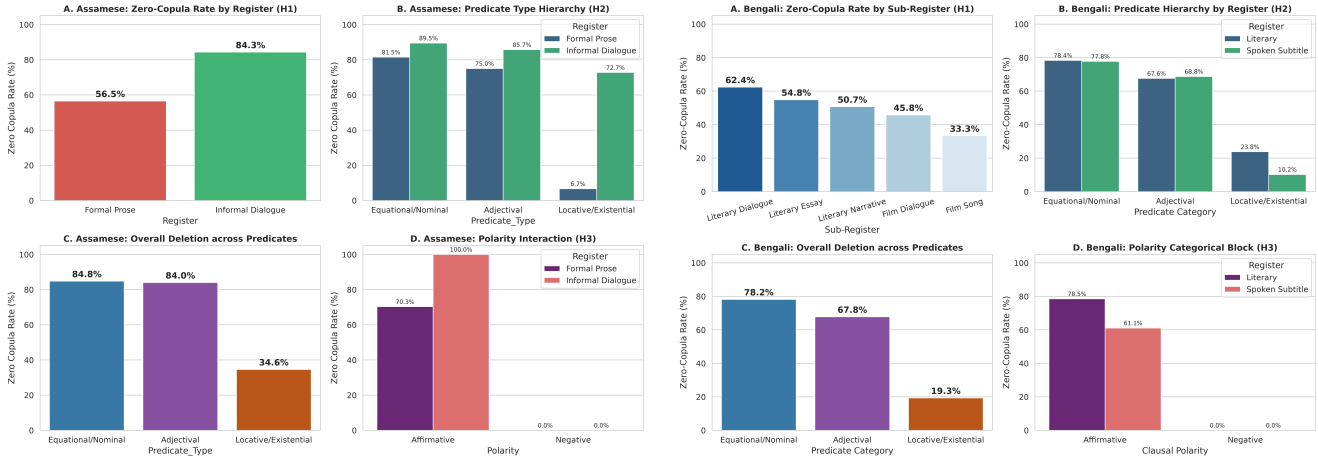


Figure 2 (Left) & Figure 3 (Right): Empirical dashboards for the Eastern Indo-Aryan (Magadhan) languages: **Assamese** (N=97, Left) and **Bengali** (N=373, Right). Both languages exhibit ceiling deletion in affirmative Equational (84.8% AS, 78.2% BN) and Adjectival (84.0% AS, 67.8% BN) predicates, strong retention of overt existential আছে (*ase/aachhe*) in Locative clauses, and 0.0% deletion in Negative clauses (নহয়/নাই and নয়/নেই). Note the sharp +66.0 pp informal register surge in Assamese Locative predicates (Figure 2, Panel B).

3.4 Telugu (South-Central Dravidian, N_{auto}=759, N_{manual}=116)

In the manually verified Telugu dataset (N=116; **Figure 4**), the overall zero-copula rate is **47.41%** (55/116):

- **H1 (Register):** Deletion rates across **Film Dialogue (42.86%, 21/49)**, **Literary Dialogue (55.56%, 20/36)**, and **Literary Narrative (45.16%, 14/31; total = 55/116)** show no significant difference between spoken film and written narrative (Fisher's exact p =

1.000). Just as in Bengali, Dravidian nominal juxtaposition is a core grammatical rule shared by modern cinema (నువ్వు నా జోడీ [Ø] 'You [are] my partner') and classic literary prose by Gurajada Apparao (గోపాలుడు సదయ హృదయుడు [Ø] 'Gopaludu [is] a kind-hearted man').

- **H2 (Predicate Hierarchy):** Telugu exhibits the strongest predicate-type conditioning of all five languages (χ^2 = 49.53, df=2, p = 1.76 × 10^{-11}, Cramér's V = **0.653**): **Equational/Nominal (79.25%, 42/53 overall, and 100.0% [42/42] in affirmative clauses)** > **Adjectival (45.83%, 11/24)** ≫ **Locative/Existential (5.13%, 2/39; total** = **55/116)**. In Telugu, 94.9% of locative and existential clauses retain overt forms of ఉండు (*undi, unnadu, unnaru, unnam*) or literary కలవు/కద్దు (*kalavu/kaddu*), and 54.2% of adjectival clauses employ adverbial -గా (*-ga*) + ఉంది (*undi*) (e.g., నిజంగానే తేలిగ్గా ఉంది 'It is really easy').
- **H3 (Polarity):** Negative clauses exhibit **0.00% deletion** (0/32 vs. 65.48%, 55/84 in affirmatives; total = 55/116, Fisher's exact p = 9.56 × 10^{-12}). Every equational/adjectival negation requires overt కాదు/కావు/కారు (*kaadu/kaavu/kaaru*), and every existential/locative negation requires overt లేదు/లేవు/లేరు (*ledu/levu/leru*).

3.5 Tamil (Southern Dravidian, N_{clean} = 106)

In Tamil (N=106 clean non-verbal clauses after filtering 15 verbal rows; **Figure 5**), the overall copula deletion rate is **70.75%** (75/106):

- **H1 × H2 Diglossic Interaction (The 58.8 pp Locative Split):** Overall deletion is significantly higher in **Informal Colloquial Dialogue (81.13%, 43/53)** than in **Formal Prose (60.38%, 32/53; total** = **75/106,** χ^2 = **4.56,** p = **0.0327,** V = **0.207)**. In both formal and informal Tamil, affirmative Nominal (84.1%) and Adjectival (85.7%) clauses overwhelmingly use zero-copula (in formal Tamil, ஆகும் *aagum* or உள்ளது *ullathu* is optional stylistic framing, and in spoken Tamil, identity clauses take no overt copula). However, inside **Locative/Existential** predicates, Tamil displays a dramatic diglossic divergence: Formal Written Tamil retains overt existential verbs (உள்ளது *ullathu,* உள்ளன *ullana,* உண்டு *undu,* இருக்கிறது *irukkirathu*) in **88.2%** of clauses (11.8% deletion, 2/17), whereas Informal Spoken Tamil drops colloquial இருக்கு/இருக்காங்க (*irukku/irukkanga*) in **70.6%** of clauses (12/17, e.g., நான் வீட்ல [Ø] 'I [am] at home'; நாளைக்கு மீட்டிங் [Ø] 'Tomorrow [there is a] meeting', attested in *Coolie*)—a **58.8 percentage-point diglossic gap** (Fisher's exact p = 0.0013).
- **H2 (Predicate Hierarchy):** Strongly confirmed (χ^2 = 21.19, df=2, p = 2.51 × 10^{-5}, Cramér's V = 0.447), with Adjectival and Equational predicates both above 84%: **Adjectival (85.71%, 24/28)** ≥ **Equational/Nominal (84.09%, 37/44)** ≫ **Locative/Existential (41.18%, 14/34; total** = **75/106)**. Notably, Tamil's zero-copula form in adjectival clauses requires morphological nominalization/pronominalization on the predicate (e.g., overt அழகா இருக்கு vs. zero அழகு or formal அழகானவை).
- **H3 (Polarity):** Negative clauses exhibit **0.00% deletion** (0/18 vs. 85.23%, 75/88 in affirmatives; total = 75/106, Fisher's exact p = 2.14 × 10^{-12}), as formal Tamil requires அல்ல (*alla*), negative-interrogative tag அல்லவா (*allava*), or இல்லை (*illai*) and colloquial Tamil generalizes இல்ல (*illa*) across all negative non-verbal predicates.

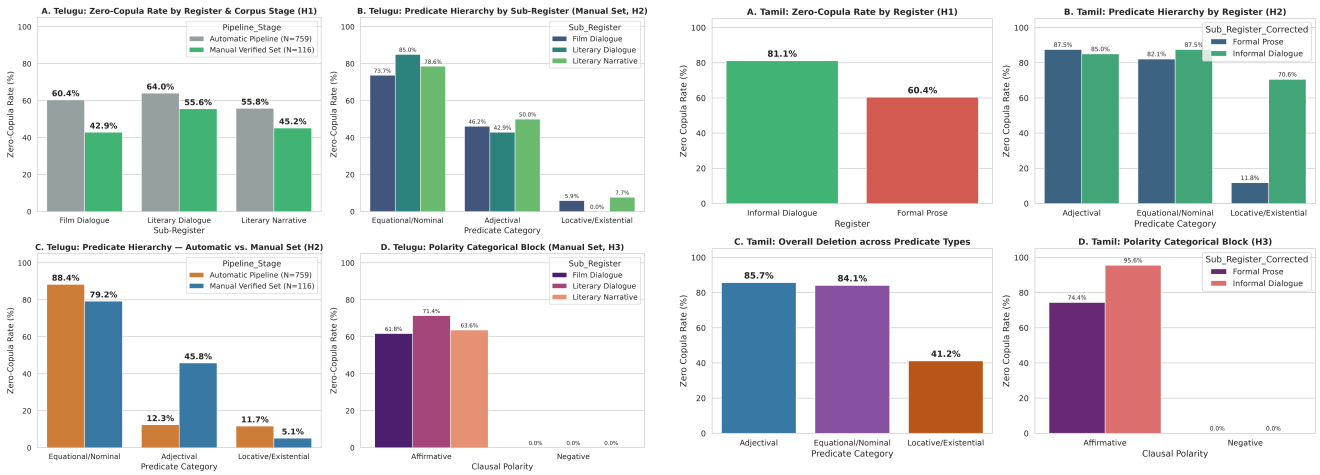


Figure 4 (Left) & Figure 5 (Right): Empirical dashboards for the Dravidian languages: **Telugu** (Dual-Track Auto N=759 vs. Manual N=116, Left) and **Tamil** (N=106, Right). Figure 4 (Panel C) illustrates how manual verification corrects the automated nominal fallback bias in Telugu, revealing the true 79.2% Equational > 45.8% Adjectival > 5.1% Locative hierarchy. Figure 5 (Panel B) highlights Tamil's 58.8 pp diglossic register gap in Locative/Existential clauses (70.6% informal deletion vs. 11.8% formal deletion).

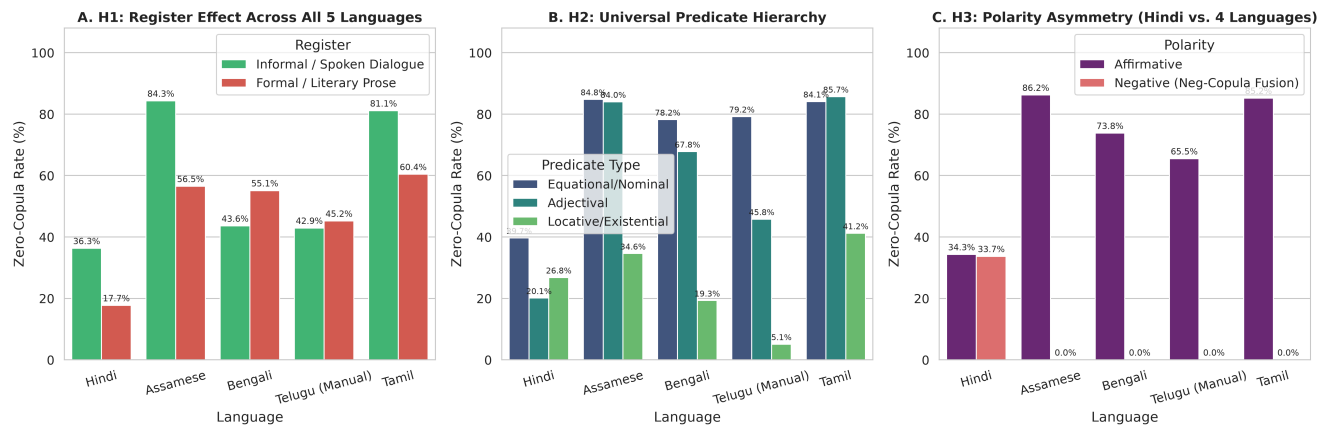


Figure 6: Master Cross-Linguistic Synthesis Dashboard comparing **H1 (Register Conditioning)**, **H2 (Universal Predicate Hierarchy)**, and **H3 (Polarity & Negation-Copula Fusion)** across all five Indian languages (N_{clean} = 3,781). Every bar matches Table 3 and Sections 3.1–3.5 to the tenth of a percent.

3.6 Cross-Linguistic Comparative Synthesis

Synthesizing all five languages in **Figure 6** and **Table 3** reveals three overarching comparative generalizations across Indo-Aryan and Dravidian:

1. **Grammaticalized vs. Stylistic Zero-Copula (H1):** A uniform register effect across all clause types emerges when the overt copula is the unmarked grammatical default—as in **Hindi** (36.3% spoken film vs. 17.7% narrative). In **Bengali, Assamese, Telugu, and Tamil**, zero-copula in affirmative nominal identity clauses is grammaticalized as the standard rule of the language, so formal prose and casual dialogue alike omit the copula (78%\–100%). However, stylistic register resurfaces sharply at the margins of grammaticalization—specifically inside **Locative/Existential clauses** in diglossic Tamil (+58.8 pp) and Assamese (+66.0 pp), where conversational routines permit dropping existential *irukku/ase* in ways disallowed by formal written prose.
2. **Predicate Conditioning Across Language Families (H2):** In all five languages without exception ($p < 10^{-4}$), Equational/Nominal predicates exhibit the highest or near-highest zero-copula rates (39.7%\–84.8%). In Assamese, Bengali, Telugu, and Tamil, Locative/Existential predicates resist deletion far more strongly than both Equational and Adjectival predicates (58.8%\–94.9% overt retention), because asserting physical location or existence in space-time requires an overt existential verb (*ase, aachhe, undi, ullathu/irukku*) to anchor figure and ground.
3. **Morphological Negation-Copula Fusion as an Absolute Barrier (H3):** Whenever negation is morphologically fused into a synthetic negative copular head (Assamese *nohoy/nai*, Bengali *noy/nei*, Telugu *kaadu/ledu*, Tamil *alla/illai*), copula deletion

drops to **0.0%**. Only when negation is expressed analytically by an independent particle (Hindi *nahin*) can the copula be freely omitted (33.7%).

4. Correlation of the Assignment with Course Concepts

Table 4 synthesizes how the empirical findings and methodological insights of our five-language copula deletion study connect directly to the core theoretical topics covered in the course.

Table 4: Explicit Mapping Between Course Topics and Our Cross-Linguistic Copula Deletion Study

Course Topic	Core Theoretical Concept Taught in Class	Direct Empirical & Computational Manifestation in Our 5-Language Copula Study
Competence vs. Performance & I-Language vs. E-Language	*I-language* is the internal mental grammar; *E-language* is external usage performance shaped by context and style.	Explains why zero-copula in affirmative identity clauses (Telugu 100%, Bengali 78.2%) and overt retention in fused negatives (100% in AS, BN, TE, TA) are categorical **I-language rules**, whereas Hindi copula drop (36.3% film vs. 17.7% narrative) is a stylistic **E-language performance** alternation.
Child Language & Function-Word Omission	Elimination of closed-class function words in favor of open-class content words (nouns, adjectives, verbs) when context is shared.	Present-tense copulas (*hai, hoy*) are closed-class function words carrying minimal information when speech is anchored in the present. Conversational speakers drop them in rapid dialogue just as children omit function words in telegraphic speech.
Usage-Based Linguistics & Entrenchment	Grammar emerges bottom-up from repeated usage; frequent constructions become entrenched in long-term memory; gradient rules vs. categorical rules.	Tracks the Indo-Aryan usage cline from gradient stylistic deletion in Central Hindi (34.2%) to entrenched, grammaticalized zero-copula in Eastern Bengali (52.0%) and Assamese (71.1%), especially in high-frequency colloquial routines.
Working Memory, Attention & Salience	Working memory has limited cue-integration capacity; attention is competitive and prioritizes salient, task-relevant cues.	Explains **H2** (locative/existential clauses require overt verbs *hai/ase/aachhe/undi/ullathu* to anchor figure-ground space-time in working memory) and **H3** (negation is maximally task-relevant and cannot be left null; Hindi *nahin* carries attentional salience so *hai* can drop).
Construal & Constructional Alternations	Speakers choose between alternative grammatical constructions to package the same situation based on perspective, prominence, and focus.	In Tamil and Assamese, overt vs. zero copula encodes construal differences: e.g., permanent identity (zero) vs. temporary role/state (Tamil -ஆ இருக்காரு) or formal definitional authority (ஆகும் *aagum* / হয় *hoy*) vs. casual reply.
Ontological Aspect of Cognition, LOTH & Relativity	Ontological distinction between conceptual attribution and real-world existence; language-specific categorization of thought.	All 5 Indian languages grammatically separate conceptual identity/attribution (zero-copula / *hoy / aagum / kaadu / alla*) from ontological existence in the world (*ase / aachhe / undi / ullathu / nai / nei / ledu / illai*).
Corpus Methodology & Human vs. Machine Learning	Statistical NLP patterns can be descriptively accurate on surface strings yet psychologically unreal without theory-guided auditing.	Directly explains the professor's observation during the presentation: raw Stanza NLP produced a spurious 59.8% literary deletion rate in Hindi due to past/action-verb "false zeros." Theory-guided auditing restored the true 17.7% rate.

5. Limitations, Conclusion & Future Work

5.1 Limitations of the Study

To contextualize our findings rigorously, we note four methodological and empirical limitations of the present study:

- **Sample Size & Corpus Structure Asymmetry:** While our Hindi (N = 3,089 main + 1,063 verification), Bengali (N = 373), and Telugu (N = 759 automated / 116 manual) datasets were extracted from continuous film transcripts and literary narratives, our Assamese (N = 97) and Tamil (N = 106) corpora were compiled as structured cross-register sentence sets pairing overt and zero realizations. Although these paired sets provide clean tests of register and predicate contrasts, larger continuous corpora for Assamese and Tamil are needed to estimate naturalistic baseline frequencies.
- **Dependency Parser & ASR Noise in Low-Resource Languages:** Off-the-shelf neural dependency parsers (Stanza) trained on standard written news corpora struggle with unpunctuated YouTube Auto-ASR transcripts (as seen in Telugu's nominal fallback bias) and literary narrative tenses (as seen in Hindi's 320 false-zero verbal clauses), requiring substantial manual linguistic auditing.

- **Intra-Language Dialectal Variation:** As demonstrated by the contrast between Phanishwar Nath Renu's regionalist Hindi fiction (*Pahalwan Ki Dholak*, 57.5% deletion) and Premchand's standard Hindi prose (*Idgah*, 12.4%), regional varieties (such as Eastern Hindi influenced by Magadhan languages, Telangana vs. Coastal Andhra Telugu, or Madurai vs. Chennai Tamil) exhibit distinct copula drop baselines that aggregate language-level models only partially capture.
- **Absence of Acoustic & Prosodic Annotations:** Subtitle and text corpora do not encode prosodic boundaries, pause duration, or pitch contours, which play a vital role in licensing zero-copula utterances and disambiguating equational juxtaposition in spoken interaction.

5.2 Conclusion

This study presented a unified computational and linguistic investigation of present-tense copula deletion across five Indian languages spanning the Indo-Aryan (Hindi, Assamese, Bengali) and Dravidian (Telugu, Tamil) families (N_{clean} = 3,781). By auditing raw NLP dependency parses against native-speaker grammatical intuition—directly resolving the counter-intuitive raw literary narrative spike noted by the professor during our presentation—we demonstrated that theory-guided linguistic filtering is essential when applying automated parsers to typologically diverse corpora. Empirically, our results establish that present-tense copula deletion in South Asian languages is governed by the interaction of three forces: (1) whether zero-copula is stylistically conditioned (Hindi) or grammaticalized in core syntax with diglossic locative extensions (Bengali, Assamese, Telugu, Tamil); (2) the semantic predicate hierarchy (where Equational/Nominal predicates favor deletion while Locative/Existential predicates retain overt existential verbs); and (3) the morphological locus of negation (analytic Hindi *nahin* permitting 33.7% drop vs. synthetic negative copulas enforcing 0.0% drop across Assamese, Bengali, Telugu, and Tamil).

5.3 Directions for Future Work

- **Expansion to Continuous Multi-Genre Corpora & Additional Language Families:** Extending automated extraction and manual linguistic auditing to large-scale continuous subtitle and novel corpora in Assamese and Tamil, as well as Marathi, Odia, Malayalam, Kannada, and Tibeto-Burman/Austroasiatic languages of India.
- **Fine-Tuning Transformer Models for Zero-Copula Detection:** Training syntax-aware multilingual transformers (such as IndicBERT and MuRIL) on our manually verified non-verbal datasets to automatically distinguish true zero-copula clauses from finite past/habitual verbal clauses without manual post-filtering.
- **Experimental Psycholinguistic Validation:** Conducting acceptability rating tasks, self-paced reading studies, and eye-tracking experiments to measure real-time cognitive processing costs when native speakers encounter overt vs. zero copulas across formal and informal registers.

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Appendix A · Sentence-Level Verification Compendium Across All Five Languages

Prepared in direct response to presentation feedback to allow full native-speaker verification of the assigned Register, Sub-Register, Predicate Type (H2), Polarity (H3), and Copula Status (0 = Overt Retained, 1 = Zero/Deleted) across Hindi, Assamese, Bengali, Telugu, and Tamil.

🔗 **Project GitHub Repository: https://github.com/nitishgupt100/language_cognition_computation**

You might visit the GitHub repository above to inspect the complete raw corpora, Stanza NLP dependency extraction scripts, interactive Jupyter notebooks (with full executed cell outputs and visualizations), and statistical analysis pipelines across all five languages.

A.1 Hindi Audited Non-Verbal Sentences (Representative Stratified Sample: $n = 64$ across all 8 Sources)

#	Source & ID	H1: Register & Sub-Register	Hindi Sentence (Devanagari) & Stanza Root	H2: Predicate Type	H3: Polarity	Assigned Copula Status
1	**Do_Bailon_Ki_Katha** *CORPUS_4097*	**Literary Narrative** *Literary*	कदाचित् सीधापन संसार के लिए उपयुक्त नहीं हैं । *Root Word: उपयुक्त*	Adjectival	Negative	Retained (Overt = 0)
2	**Do_Bailon_Ki_Katha** *CORPUS_4103*	**Literary Narrative** *Literary*	जापान की मिशाल सामने हैं । *Root Word: सामने*	Equational/Nominal	Affirmative	Retained (Overt = 0)
3	**Do_Bailon_Ki_Katha** *CORPUS_4165*	**Literary Narrative** *Literary*	यहाँ भी किसी सज्जन का वास हैं । *Root Word: वास*	Equational/Nominal	Affirmative	Retained (Overt = 0)
4	**Do_Bailon_Ki_Katha** *CORPUS_4211*	**Literary Narrative** *Literary*	कितनी भयंकर सूरत हैं । *Root Word: सूरत*	Equational/Nominal	Affirmative	Retained (Overt = 0)
5	**Do_Bailon_Ki_Katha** *CORPUS_4311*	**Literary Dialogue** *Literary*	'तुम्हारे बैल कैसे ? *Root Word: बैल*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
6	**Do_Bailon_Ki_Katha** *CORPUS_4141*	**Literary Narrative** *Literary*	न कोई चिकनाहट, न कोई रस । *Root Word: चिकनाहट*	Equational/Nominal	Negative	Deleted (Zero = 1)
7	**Do_Bailon_Ki_Katha** *CORPUS_4157*	**Literary Narrative** *Literary*	दोनों के हाथ में लाठियाँ हैं । *Root Word: लाठियाँ*	Locative/Existential	Affirmative	Retained (Overt = 0)
8	**Do_Bailon_Ki_Katha** *CORPUS_4256*	**Literary Dialogue** *Literary*	' 'कुछ परवाह नहीं । *Root Word: परवाह*	Locative/Existential	Negative	Deleted (Zero = 1)
9	**Idgah** *CORPUS_3628*	**Literary Narrative** *Literary*	लड़के सबसे ज़्यादा प्रसन्न हैं। *Root Word: प्रसन्न*	Adjectival	Affirmative	Retained (Overt = 0)
10	**Idgah** *CORPUS_3619*	**Literary Narrative** *Literary*	कितना मनोहर, कितना सुहावना प्रभाव है। *Root Word: प्रभाव*	Equational/Nominal	Affirmative	Retained (Overt = 0)
11	**Idgah** *CORPUS_3753*	**Literary Dialogue** *Literary*	'कहाँ पचास, कहाँ एक सौ। *Root Word: एक*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
12	**Idgah** *CORPUS_3827*	**Literary Narrative** *Literary*	खिलौने से क्या फायदा? *Root Word: फायदा*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
13	**Idgah** *CORPUS_3667*	**Literary Narrative** *Literary*	जूते भी तो नहीं हैं। *Root Word: जूते*	Equational/Nominal	Negative	Retained (Overt = 0)
14	**Idgah** *CORPUS_3620*	**Literary Narrative** *Literary*	वृक्षों पर अजीब हरियाली है, खेतों में कुछ अजीब रौनक है, आसमान पर कुछ अजीब लालिमा है। *Root Word: हरियाली*	Locative/Existential	Affirmative	Retained (Overt = 0)

#	Source & ID	H1: Register & Sub-Register	Hindi Sentence (Devanagari) & Stanza Root	H2: Predicate Type	H3: Polarity	Assigned Copula Status
15	**Idgah** *CORPUS_3758*	**Literary Narrative** *Literary*	कोई इक्के-ताँगे पर सवार, कोई मोटर पर, सभी इत्र में बसे, सभी के दिलों में उमंग। *Root Word: उमंग।*	Locative/Existential	Affirmative	Deleted (Zero = 1)
16	**Idgah** *CORPUS_3966*	**Literary Dialogue** *Literary*	' 'कै पैसे में? *Root Word: पैसे*	Locative/Existential	Affirmative	Deleted (Zero = 1)
17	**Pahlwan_Ki_Dholak** *CORPUS_3975*	**Literary Narrative** *Literary*	जाड़े का दिन। *Root Word: दिन।*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
18	**Pahlwan_Ki_Dholak** *CORPUS_3978*	**Literary Narrative** *Literary*	अन्धेरा और निस्तबधता ! *Root Word: अन्धेरा*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
19	**Pahlwan_Ki_Dholak** *CORPUS_3987*	**Literary Narrative** *Literary*	लुट्टन सिंह पहलवान ! *Root Word: पहलवान*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
20	**Pahlwan_Ki_Dholak** *CORPUS_3988*	**Literary Narrative** *Literary*	किन्तु उसके होल इण्डिया की सीमा शायद एक ज़िले की सीमा के बराबर ही हो। *Root Word: हो*	Equational/Nominal	Affirmative	Retained (Overt = 0)
21	**Pahlwan_Ki_Dholak** *CORPUS_4013*	**Literary Dialogue** *Literary*	"वाह रे मिट्टी के शेर! *Root Word: शेर!*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
22	**Pahlwan_Ki_Dholak** *CORPUS_4008*	**Literary Dialogue** *Literary*	हंसी—खेल नहीं—शेर का बच्चा है... बच्चू! *Root Word: बच्चू*	Equational/Nominal	Negative	Retained (Overt = 0)
23	**Pahlwan_Ki_Dholak** *CORPUS_4054*	**Literary Narrative** *Literary*	हां, मेरा गुरू कोई पहलवान नहीं, यही ढोल है, समझे! *Root Word: पहलवान*	Equational/Nominal	Negative	Retained (Overt = 0)
24	**Pahlwan_Ki_Dholak** *CORPUS_3979*	**Literary Narrative** *Literary*	पृथ्वी पर कहीं प्रकाश का नाम नहीं। *Root Word: नाम*	Locative/Existential	Negative	Deleted (Zero = 1)
25	**Poos_Ki_Raat** *VERIFY_1652*	**Literary Narrative** *Literary*	फसल तैयार है। *Root Word: तैयार*	Adjectival	Affirmative	Retained (Overt = 0)
26	**Poos_Ki_Raat** *VERIFY_1576*	**Literary Narrative** *Literary*	अच्छी खेती है! *Root Word: खेती*	Equational/Nominal	Affirmative	Retained (Overt = 0)
27	**Poos_Ki_Raat** *VERIFY_1587*	**Literary Narrative** *Literary*	यह खेती का मज़ा है! *Root Word: मज़ा*	Equational/Nominal	Affirmative	Retained (Overt = 0)
28	**Poos_Ki_Raat** *VERIFY_1589*	**Literary Narrative** *Literary*	मोटे-मोटे गद्दे, लिहाफ़-कम्मल। *Root Word: गद्दे*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
29	**Poos_Ki_Raat** *VERIFY_1617*	**Literary Narrative** *Literary*	अभी तो बहुत रात है। *Root Word: रात*	Equational/Nominal	Affirmative	Retained (Overt = 0)
30	**Poos_Ki_Raat** *VERIFY_1609*	**Literary Narrative** *Literary*	अभी पहर से ऊपर रात है। *Root Word: रात*	Locative/Existential	Affirmative	Retained (Overt = 0)
31	**3_Idiots** *CORPUS_0021*	**Film Dialogue** *Spoken Subtitle*	हाँ ठीक ठीक है, ओके। *Root Word: ठीक*	Adjectival	Affirmative	Retained (Overt = 0)

#	Source & ID	H1: Register & Sub-Register	Hindi Sentence (Devanagari) & Stanza Root	H2: Predicate Type	H3: Polarity	Assigned Copula Status
32	**3_Idiots** *CORPUS_0004*	**Film Dialogue** *Spoken Subtitle*	नहीं मैं बिल्कुल ठीक हूँ। *Root Word: ठीक*	Adjectival	Negative	Retained (Overt = 0)
33	**3_Idiots** *CORPUS_0007*	**Film Dialogue** *Spoken Subtitle*	तुस्सी मि. ढिल्लों? *Root Word: ढिल्लों*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
34	**3_Idiots** *CORPUS_0027*	**Film Dialogue** *Spoken Subtitle*	सेम सरनेम यार -- ढिल्लों। *Root Word: ढिल्लों*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
35	**3_Idiots** *CORPUS_0028*	**Film Dialogue** *Spoken Subtitle*	हे चतुर, रैंचो कहाँ है? *Root Word: कहाँ*	Equational/Nominal	Affirmative	Retained (Overt = 0)
36	**3_Idiots** *CORPUS_0055*	**Film Dialogue** *Spoken Subtitle*	मैं मन मोहन। *Root Word: मन*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
37	**3_Idiots** *CORPUS_0058*	**Film Dialogue** *Spoken Subtitle*	फिक्स्ड रेट है। *Root Word: रेट*	Equational/Nominal	Affirmative	Retained (Overt = 0)
38	**3_Idiots** *CORPUS_0061*	**Film Dialogue** *Spoken Subtitle*	ये मेगाबाइट और ये इनकी माँ, गीगाबाइट। *Root Word: मेगाबाइट*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
39	**3_Idiots** *CORPUS_0138*	**Film Dialogue** *Spoken Subtitle*	स्कूल नहीं है तेरा? *Root Word: स्कूल*	Equational/Nominal	Negative	Retained (Overt = 0)
40	**Mirzapur** *VERIFY_0055*	**Film Dialogue** *Spoken Subtitle*	- बबलू भैया... आप तो समझदार हैं। *Root Word: समझदार*	Adjectival	Affirmative	Retained (Overt = 0)
41	**Mirzapur** *VERIFY_0003*	**Film Dialogue** *Spoken Subtitle*	और इस पावर का प्रतीक है, ये गद्दी। *Root Word: प्रतीक*	Equational/Nominal	Affirmative	Retained (Overt = 0)
42	**Mirzapur** *VERIFY_0008*	**Film Dialogue** *Spoken Subtitle*	उनकी तबीयत थोड़ा डाउन है, भैया जी। *Root Word: डाउन*	Equational/Nominal	Affirmative	Retained (Overt = 0)
43	**Mirzapur** *VERIFY_0098*	**Film Dialogue** *Spoken Subtitle*	समझ नहीं है? *Root Word: समझ*	Equational/Nominal	Negative	Retained (Overt = 0)
44	**Mirzapur** *VERIFY_0187*	**Film Dialogue** *Spoken Subtitle*	और नौकरों के तो बिल्कुल नहीं। *Root Word: बिल्कुल*	Equational/Nominal	Negative	Deleted (Zero = 1)
45	**Mirzapur** *VERIFY_0014*	**Film Dialogue** *Spoken Subtitle*	स्पेशल मेहमान हैं आप, जैसवाल जी। *Root Word: मेहमान*	Locative/Existential	Affirmative	Retained (Overt = 0)

#	Source & ID	H1: Register & Sub-Register	Hindi Sentence (Devanagari) & Stanza Root	H2: Predicate Type	H3: Polarity	Assigned Copula Status
46	**Mirzapur** *VERIFY_0028*	**Film Dialogue** *Spoken Subtitle*	क्योंकी चुनाव सर पे है। *Root Word: सर*	Locative/Existential	Affirmative	Retained (Overt = 0)
47	**Mirzapur** *VERIFY_0078*	**Film Dialogue** *Spoken Subtitle*	सात-आठ दाईं तरफ़, बाकि बाईं तरफ़। *Root Word: दाईं*	Locative/Existential	Affirmative	Deleted (Zero = 1)
48	**Secret_Superstar** *CORPUS_1384*	**Film Dialogue** *Spoken Subtitle*	बहुत अच्छा है। *Root Word: अच्छा*	Adjectival	Affirmative	Retained (Overt = 0)
49	**Secret_Superstar** *CORPUS_1427*	**Film Dialogue** *Spoken Subtitle*	पता नहीं, बहुत मुश्किल है। *Root Word: मुश्किल*	Adjectival	Negative	Retained (Overt = 0)
50	**Secret_Superstar** *CORPUS_1374*	**Film Dialogue** *Spoken Subtitle*	चेस्ट - 54। चेस्ट - 19 सी। *Root Word: सी।*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
51	**Secret_Superstar** *CORPUS_1383*	**Film Dialogue** *Spoken Subtitle*	इंसिया, नया गाना है? *Root Word: गाना*	Equational/Nominal	Affirmative	Retained (Overt = 0)
52	**Secret_Superstar** *CORPUS_1386*	**Film Dialogue** *Spoken Subtitle*	ये इंटर-स्कूल कल्चरल कॉम्पिटिशन के पैंफलेट हैं। *Root Word: पैंफलेट*	Equational/Nominal	Affirmative	Retained (Overt = 0)
53	**Secret_Superstar** *CORPUS_1387*	**Film Dialogue** *Spoken Subtitle*	सोलो डांसिंग, ग्रुप डांसिंग, ब्यूटी कॉन्टेस्ट और सिंगिंग भी। *Root Word: डांसिंग*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
54	**Secret_Superstar** *CORPUS_1398*	**Film Dialogue** *Spoken Subtitle*	एक भी नहीं? *Root Word: एक*	Equational/Nominal	Negative	Deleted (Zero = 1)
55	**Secret_Superstar** *CORPUS_1399*	**Film Dialogue** *Spoken Subtitle*	न सलाम, न दुआ? *Root Word: दुआ*	Equational/Nominal	Negative	Deleted (Zero = 1)
56	**Secret_Superstar** *CORPUS_1459*	**Film Dialogue** *Spoken Subtitle*	शर्म ही नहीं है। *Root Word: शर्म*	Equational/Nominal	Negative	Retained (Overt = 0)
57	**Taare_Zameen_Par** *CORPUS_2789*	**Film Dialogue** *Spoken Subtitle*	बीजगणित, ज्योमेट्री, भौतिक, रसायन, जीव विज्ञान, इतिहास, अंग्रेजी, भूगोल... बहुत अच्छे! *Root Word: अच्छे*	Adjectival	Affirmative	Deleted (Zero = 1)
58	**Taare_Zameen_Par** *CORPUS_2895*	**Film Dialogue** *Spoken Subtitle*	हर शब्द की वर्तनी गलत है! *Root Word: गलत*	Adjectival	Affirmative	Retained (Overt = 0)
59	**Taare_Zameen_Par** *CORPUS_2787*	**Film Dialogue** *Spoken Subtitle*	ड्रामा अभ्यास कैंसल्ड मॉम! *Root Word: मॉम*	Equational/Nominal	Affirmative	Deleted (Zero = 1)

#	Source & ID	H1: Register & Sub-Register	Hindi Sentence (Devanagari) & Stanza Root	H2: Predicate Type	H3: Polarity	Assigned Copula Status
60	**Taare_Zameen_Par** *CORPUS_2809*	**Film Dialogue** *Spoken Subtitle*	बच्चा है या राक्षस! *Root Word: बच्चा*	Equational/Nominal	Affirmative	Retained (Overt = 0)
61	**Taare_Zameen_Par** *CORPUS_2857*	**Film Dialogue** *Spoken Subtitle*	जूते पॉलिश नहीं हैं! *Root Word: पॉलिश*	Equational/Nominal	Negative	Retained (Overt = 0)
62	**Taare_Zameen_Par** *CORPUS_2860*	**Film Dialogue** *Spoken Subtitle*	पेज 38, चैप्टर 4, पैरा 3। ईशान, क्या तुम्हारा ध्यान यहाँ है? *Root Word: यहाँ*	Locative/Existential	Affirmative	Retained (Overt = 0)
63	**Taare_Zameen_Par** *CORPUS_3084*	**Film Dialogue** *Spoken Subtitle*	नज़र इधर, ब्लैकबोर्ड पे! *Root Word: इधर*	Locative/Existential	Affirmative	Deleted (Zero = 1)
64	**Taare_Zameen_Par** *CORPUS_2909*	**Film Dialogue** *Spoken Subtitle*	पापा भी घर पे नहीं हैं। *Root Word: घर*	Locative/Existential	Negative	Retained (Overt = 0)

A.2 Diagnostic Examples of "False-Zero" Verbal/Tense Sentences Filtered Out During Hindi Auditing (n = 10)

These examples from `Cross_register_audited_full.csv` illustrate why the raw unfiltered literary narrative deletion rate was artificially inflated (59.8%): each sentence lacks present-tense है (*hai*) because it contains a finite past, habitual, or future action verb, **not** a deleted copula. Our audit pipeline flagged `Is_Valid_NonVerbal = False` and excluded them from the clean analysis (17.7%).

#	Literary Source & ID	Raw Literary Sentence (Devanagari)	Raw Unfiltered Tag	Audit Classification Flag	Audit Decision
1	**Idgah** *CORPUS_3639*	इन्हीं अनगिनती पैसों में अनगिनती चीजें लाएँगें—खिलौने, मिठाइयाँ, बिगुल, गेंद और जाने क्या-क्या। *Root Word: लाएँगें*	Raw: Copula_Status=0	Future/Subjunctive Suffix	Excluded (Verbal Clause)
2	**Idgah** *CORPUS_3644*	बहुत-सी थैलियाँ लेकर आएँगे। *Root Word: आएँगे*	Raw: Copula_Status=0	Future/Subjunctive Suffix	Excluded (Verbal Clause)
3	**Idgah** *CORPUS_3647*	जब उसके अब्बाजान थैलियाँ और अम्मीजान नियमतें लेकर आएँगी, तो वह दिल से अरमान निकाल लेगा। *Root Word: निकाल*	Raw: Copula_Status=0	Future/Subjunctive Suffix	Excluded (Verbal Clause)
4	**Idgah** *CORPUS_3651*	आज आबिद होता, तो क्या इसी तरह ईद आती ओर चली जाती! *Root Word: चली*	Raw: Copula_Status=0	Past/Action/Imperative Verb	Excluded (Verbal Clause)
5	**Idgah** *CORPUS_3656*	विपत्ति अपना सारा दलबल लेकर आये, हामिद की आनंद-भरी चितबन उसका विध्वसं कर देगी। *Root Word: आये*	Raw: Copula_Status=0	Past/Action/Imperative Verb	Excluded (Verbal Clause)
6	**Idgah** *CORPUS_3666*	पैर में छाले पड़ जाएँगे। *Root Word: पड़*	Raw: Copula_Status=0	Future/Subjunctive Suffix	Excluded (Verbal Clause)
7	**Idgah** *CORPUS_3668*	पैसे होते तो लौटते-लोटते सब सामग्री जमा करके चटपट बना लेती। *Root Word: बना*	Raw: Copula_Status=0	Verbal Root	Excluded (Verbal Clause)
8	**Idgah** *CORPUS_3669*	यहाँ तो घंटों चीजें जमा करते लगेंगे। *Root Word: करते*	Raw: Copula_Status=0	Verbal Root	Excluded (Verbal Clause)
9	**Idgah** *CORPUS_3670*	माँगे का ही तो भरोसा ठहरा। *Root Word: ठहरा।*	Raw: Copula_Status=0	Past/Action/Imperative Verb	Excluded (Verbal Clause)

#	Literary Source & ID	Raw Literary Sentence (Devanagari)	Raw Unfiltered Tag	Audit Classification Flag	Audit Decision
10	**Idgah** *CORPUS_3683*	फिर किसी पेड़ के नींचे खड़े होकर साथ वालों का इंतजार करते। *Root Word: करते*	Raw: Copula_Status=0	**Verbal Root**	Excluded (Verbal Clause)

A.3 Complete Assamese Audited Dataset (All N = 97 Valid Non-Verbal Clauses)

#	H1: Register	Overt Copula Form (Assamese)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status
1	**Formal Prose** *Tag: Nominal*	এইবোৰেই কুসংস্কাৰ হয়।	এইবোৰেই কুসংস্কাৰ। *"These are superstitions."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
2	**Formal Prose** *Tag: Nominal*	সংস্কাৰ এক জীৱন-পদ্ধতি হয়।	সংস্কাৰ এক জীৱন-পদ্ধতি। *"Reform is a way of life."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
3	**Formal Prose** *Tag: Nominal*	ডাইনী এক সামাজিক ব্যাধি হয়।	ডাইনী এক সামাজিক ব্যাধি। *"Witchcraft is a social malady."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
4	**Formal Prose** *Tag: Nominal*	ডাইনী এক ভ্ৰান্ত বিশ্বাস হয়।	ডাইনী এক ভ্ৰান্ত বিশ্বা *"Witchcraft is a false belief."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
5	**Formal Prose** *Tag: Nominal*	নম্ৰতা সজ ব্যৱহাৰৰ অনুপান হয়।	নম্ৰতা সজ ব্যৱহাৰৰ অনুপান। *"Humility is an essential supplement to good behaviour."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
6	**Formal Prose** *Tag: Nominal*	নম্ৰতা এটা মনৰ গুণ হয়।	নম্ৰতা এটা মনৰ গুণ। *"Humility is a quality of the mind."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
7	**Formal Prose** *Tag: Nominal*	সময়ৰ প্ৰকৃত যন্ত্ৰ-পাতি কৰ্ম হয়।	সময়ৰ প্ৰকৃত যন্ত্ৰ-পাতি কৰ্ম। *"Work is the actual instrument of time."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
8	**Formal Prose** *Tag: Nominal*	সময় অমূল্য ধন হয়।	সময় অমূল্য ধন। *"Time is invaluable wealth."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
9	**Formal Prose** *Tag: Nominal/identify*	গাঁওখনৰ নাম পিপলান্ত্ৰি হয়।	গাঁওখনৰ নাম পিপলান্ত্ৰি। *"The name of the village is Piplantri."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
10	**Formal Prose** *Tag: Nominal*	নম্ৰতা ডাঙৰ আশয়ৰ চিন হয়।	নম্ৰতা ডাঙৰ আশয়ৰ চিন। *"Humility is a sign of noble character."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
11	**Formal Prose** *Tag: Adjectival*	এই শ্ৰেণীৰ মানুহৰ স্বভাৱ নিচেই পাতল হয়।	এই শ্ৰেণীৰ মানুহৰ স্বভাৱ নিচেই পাতল। *"The nature of this category of people is very frivolous."*	Adjectival	Affirmative	Deleted (Zero = 1)
12	**Formal Prose** *Tag: Adjectival/Nominal*	অন্ধবিশ্বাস আৰু কুসংস্কাৰ পৃথিৱীৰ আটাইতকৈ গধুৰ বোজা হয়।	অন্ধবিশ্বাস আৰু কুসংস্কাৰ পৃথিৱীৰ আটাইতকৈ গধুৰ বোজা। *"Superstition and blind beliefs are the heaviest burdens on the world."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
13	**Formal Prose** *Tag: Adjectival/Nominal*	সাম্প্ৰতিক অসমৰ কাৰণে ই এক জ্বলন্ত সামাজিক সমস্যা হয়।	সাম্প্ৰতিক অসমৰ কাৰণে ই এক জ্বলন্ত সামাজিক সমস্যা। *"It is a burning social problem for present-day Assam."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
14	**Formal Prose** *Tag: Negative/Existential*	আচলতে ডাইনী বুলি একো নাই।	—— *"In reality, there is no such thing as a witch."*	Locative/Existential	Negative	Retained (Overt = 0)

#	H1: Register	Overt Copula Form (Assamese)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status
15	**Formal Prose** *Tag: Existential*	পৃথিৱীত নানা ৰকমৰ মানুহ আছে।	পৃথিৱীত নানা ৰকমৰ মানুহ। *"There are different kinds of people in the world."*	Locative/Existential	Affirmative	Deleted (Zero = 1)
16	**Formal Prose** *Tag: Locative/Existential*	এই দেশৰ বাবে মই জীয়াই আছোঁ।	—— *"I am alive for this country."*	Locative/Existential	Affirmative	Retained (Overt = 0)
17	**Formal Prose** *Tag: Locative/Existential*	অনৈক্যৰ মাজত ঐক্য হৈ, বিৰোধৰ মাজত সংহতিৰ সম্ভাৱনা হৈ মই আছোঁ।	—— *"I exist as unity amid diversity and as the possibility of harmony amid conflict."*	Locative/Existential	Affirmative	Retained (Overt = 0)
18	**Formal Prose** *Tag: Existential*	সময়বিভাগৰ গুৰিতে এনে একোটা স্মৰণীয় ঘটনা আছে।	—— *"There is a memorable event at the origin of each division of time."*	Locative/Existential	Affirmative	Retained (Overt = 0)
19	**Formal Prose** *Tag: Existential*	গাঁওখনৰ পুনৰুজ্জীৱনৰ আঁৰত এজন ব্যক্তিৰ অক্লান্ত পৰিশ্ৰম আছে।	—— *"Behind the revival of the village lies the tireless effort of one person."*	Locative/Existential	Affirmative	Retained (Overt = 0)
20	**Formal Prose** *Tag: Existential*	আমাৰ মাজত আৰু এটা প্ৰৱণতা আছে।	—— *"There is another tendency among us."*	Locative/Existential	Affirmative	Retained (Overt = 0)
21	**Formal Prose** *Tag: Negative/Locative*	সময়ৰ জোখ ঘড়ীৰ কাঁইটকেইডালত নাই।	—— *"The measurement of time is not in the hands of a clock."*	Locative/Existential	Negative	Retained (Overt = 0)
22	**Formal Prose** *Tag: Interrogative/nominal*	এয়া আশ্চৰ্যৰ কথা নহয়নে ?	—— *"Isn't this a matter of wonder?"*	Equational/Nominal	Negative	Retained (Overt = 0)
23	**Formal Prose** *Tag: Interrogative/nominal*	সময় অমূল্য ধন নহয়নে?	—— *"Isn't time invaluable wealth?"*	Equational/Nominal	Negative	Retained (Overt = 0)
24	**Formal Prose** *Tag: Interrogative/nominal*	নম্ৰতা এটা মনৰ গুণ নহয়নে?	—— *"Isn't humility a quality of the mind?"*	Equational/Nominal	Negative	Retained (Overt = 0)
25	**Formal Prose** *Tag: Interrogative/nominal*	ডাইনী এক ভ্ৰান্ত বিশ্বাস নহয়নে?	—— *"Isn't witchcraft a false belief?"*	Equational/Nominal	Negative	Retained (Overt = 0)
26	**Formal Prose** *Tag: Nominal/classification*	সময়ৰ ডাঙৰ ভাগবিলাক যুগ, কল্প আৰু শতাব্দী হয়।	সময়ৰ ডাঙৰ ভাগবিলাক যুগ, কল্প আৰু শতাব্দী। *"The major divisions of time are ages, aeons and centuries."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
27	**Formal Prose** *Tag: Nominal/classification*	মজলীয়া ভাগবিলাক ঋতু, বছৰ আৰু মাহ হয়।	মজলীয়া ভাগবিলাক ঋতু, বছৰ আৰু মাহ। *"The intermediate divisions are seasons, years and months."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
28	**Formal Prose** *Tag: Adjectival*	সময়ৰ এই ভাগবিলাক বিশেষ বিশেষ লক্ষণাক্ৰান্ত হয়।	সময়ৰ এই ভাগবিলাক বিশেষ বিশেষ লক্ষণাক্ৰান্ত। *"These divisions of time have particular characteristics."*	Adjectival	Affirmative	Deleted (Zero = 1)

#	H1: Register	Overt Copula Form (Assamese)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status
29	**Formal Prose** *Tag: Nominal*	এই যুগবিলাকৰ পৰিৱর্তনেই কল্পান্তৰ হয়।	এই যুগবিলাকৰ পৰিৱর্তনেই কল্পান্তৰ। *"The transition of these ages is called a change of aeon."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
30	**Formal Prose** *Tag: Nominal*	মানুহৰ এক কল্প ব্ৰহ্মাৰ এদিন আৰু এৰাতি হয়।	মানুহৰ এক কল্প ব্ৰহ্মাৰ এদিন আৰু এৰাতি। *"One human aeon is equivalent to one day and one night of Brahma."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
31	**Formal Prose** *Tag: Nominal*	এইসকল সন্মানৰ পাত্র হয়।	এইসকল সন্মানৰ পাত্র। *"These are worthy of respect."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
32	**Formal Prose** *Tag: Nominal*	মানুহৰ সেৱা মানুহৰ পূজা হয়।	মানুহৰ সেৱা মানুহৰ পূজা। *"Service to humanity is worship of humanity"*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
33	**Formal Prose** *Tag: Nominal*	সুখ-দুখ মনৰেই কাৰণ হয়।	সুখ-দুখ মনৰেই কাৰণ। *"Happiness and sorrow originate in the mind."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
34	**Formal Prose** *Tag: Nominal*	মানৱ জীৱন ধর্মময় জীৱন হয়।	মানৱ জীৱন ধর্মময় জীৱন *"Human life is a life of righteousness."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
35	**Formal Prose** *Tag: Defenitional*	সন্মান মানে হৈছে গুণ, পদমর্যাদা, জ্যেষ্ঠতা আদিৰ ভিত্তিত লাভ কৰা শ্রদ্ধা বা আদৰ।	—— *"Respect means the regard or esteem received on the basis of qualities, status and seniority."*	Equational/Nominal	Affirmative	Retained (Overt = 0)
36	**Formal Prose** *Tag: Existential*	পৃথিৱীত মানুহৰ মান্য লোক আছে।	—— *"There are respected people in the world."*	Locative/Existential	Affirmative	Retained (Overt = 0)
37	**Formal Prose** *Tag: Exiatential*	সকলোৰে আদি আছে।	—— *"Everything has a beginning."*	Locative/Existential	Affirmative	Retained (Overt = 0)
38	**Formal Prose** *Tag: Existential*	সকলোৰে অন্ত আছ।	—— *"Everything has an end."*	Locative/Existential	Affirmative	Retained (Overt = 0)
39	**Formal Prose** *Tag: Nominal*	পৃথিৱী আমাৰ ঘৰ।	পৃথিৱী আমাৰ ঘৰ। *"The world is our home."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
40	**Formal Prose** *Tag: Nominal*	মানুহৰ সেৱা পবিত্র ব্রত।	মানুহৰ সেৱা পবিত্র ব্রত। *"Service to humanity is a sacred vow."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
41	**Formal Prose** *Tag: Adjectival*	সন্মান সকলোৰে বাবে প্রয়োজনীয়।	সন্মান সকলোৰে বাবে প্রয়োজনীয়। *"Respect is necessary for everyone."*	Adjectival	Affirmative	Deleted (Zero = 1)
42	**Formal Prose** *Tag: Existential*	আমাৰ ব্যক্তিগত আৰু সামাজিক উভয় জীৱনতে সন্মানৰ প্রয়োজন আছে।	—— *"Respect is necessary in both our personal and social lives."*	Locative/Existential	Affirmative	Retained (Overt = 0)
43	**Formal Prose** *Tag: Locative/Existential*	মানুহৰ মনত সুখ-দুখ থাকে।	—— *"Happiness and sorrow exist in the human mind."*	Locative/Existential	Affirmative	Retained (Overt = 0)

#	H1: Register	Overt Copula Form (Assamese)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status
44	**Formal Prose** *Tag: Negative/Existential*	মানুহৰ জীৱনত সুখ-শান্তি নাথাকে।	—— *"There is no happiness and peace in people's lives."*	Locative/Existential	Negative	Retained (Overt = 0)
45	**Formal Prose** *Tag: Negative/Possesive*	যাৰ তেওঁৰ শ্ৰদ্ধা-ভক্তি নাই, তাক নাস্তিক বুলি ধৰা হয়।	—— *"One who has no respect or devotion towards Him is considered an atheist."*	Locative/Existential	Negative	Retained (Overt = 0)
46	**Formal Prose** *Tag: Negative/adjectival*	সন্মানৰ অভাৱ সমাজৰ বাবে ভাল নহয়।	—— *"Lack of respect is not good for society."*	Adjectival	Negative	Retained (Overt = 0)
47	**Informal Dialogue** *Tag: Nominal*	তাই মোৰ ভণ্টি হয়।	তাই মোৰ ভণ্টি। *"She is my younger sister."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
48	**Informal Dialogue** *Tag: Nominal*	এইখন মোৰ কিতাপ হয়।	এইখন মোৰ কিতাপ *"This is my book."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
49	**Informal Dialogue** *Tag: Nominal*	এইটো এটা ডাঙৰ সমস্যা হয়।	এইটো এটা ডাঙৰ সমস্যা। *"This is a big issue"*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
50	**Informal Dialogue** *Tag: Adjectival*	বতৰটো বৰ গৰম হয়।	বতৰটো বৰ গৰম। *"The weather is very hot"*	Adjectival	Affirmative	Deleted (Zero = 1)
51	**Informal Dialogue** *Tag: Adjectival*	এই কামটো সহজ হয়।	এই কামটো সহজ। *"This task is easy"*	Adjectival	Affirmative	Deleted (Zero = 1)
52	**Informal Dialogue** *Tag: Adjectival*	মানুহজন বৰ মৰমিয়াল হয়।	মানুহজন বৰ মৰমিয়াল। *"The man is very affectionate"*	Adjectival	Affirmative	Deleted (Zero = 1)
53	**Informal Dialogue** *Tag: Adjectival*	ঠাইখন বৰ ধুনীয়া হয়।	ঠাইখন বৰ ধুনীয়া *"The place is very beautiful"*	Adjectival	Affirmative	Deleted (Zero = 1)
54	**Informal Dialogue** *Tag: Adjectival*	তাই বৰ শান্ত হয়।	তাই বৰ শান্ত। *"She is very calm"*	Adjectival	Affirmative	Deleted (Zero = 1)
55	**Informal Dialogue** *Tag: Adjevtival*	এই চিনেমাখন বৰ ভাল হয়।	এই চিনেমাখন বৰ ভাল। *"This cinema is very nice"*	Adjectival	Affirmative	Deleted (Zero = 1)
56	**Informal Dialogue** *Tag: Locative*	মই ঘৰত আছোঁ	মই ঘৰত। *"I am in home"*	Locative/Existential	Affirmative	Deleted (Zero = 1)
57	**Informal Dialogue** *Tag: Locative*	মা পাকঘৰত আছে।	মা পাকঘৰত। *"Mother is in kitchen"*	Locative/Existential	Affirmative	Deleted (Zero = 1)
58	**Informal Dialogue** *Tag: Locative*	তাই এতিয়া ঘৰত আছ	তাই এতিয়া ঘৰত। *"She is in home now"*	Locative/Existential	Affirmative	Deleted (Zero = 1)
59	**Informal Dialogue** *Tag: Existential*	বাহিৰত মানুহ আছে।	বাহিৰত মানুহ। *"There are people outside"*	Locative/Existential	Affirmative	Deleted (Zero = 1)
60	**Informal Dialogue** *Tag: Negative Nominal*	তাই ডাক্তৰ নহয়।	———— *"She is not a doctor"*	Equational/Nominal	Negative	Retained (Overt = 0)
61	**Informal Dialogue** *Tag: Negative Adjectival*	কথাটো সঁচা নহয়।	———— *"This statement is not true"*	Adjectival	Negative	Retained (Overt = 0)
62	**Informal Dialogue** *Tag: Negative Adjectival*	আজি বতৰ ভাল নহয়।	———— *"The weather is not good today"*	Adjectival	Negative	Retained (Overt = 0)
63	**Informal Dialogue** *Tag: Interrogative Nominal*	এইটো তোমাৰ ফোন হয় নেকি?	এইটো তোমাৰ ফোন নেকি? *"Is this your phone?"*	Equational/Nominal	Affirmative	Deleted (Zero = 1)

#	H1: Register	Overt Copula Form (Assamese)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status
64	**Informal Dialogue** *Tag: Interrogative Adjectival*	তাই বৰ ধুনীয়া হয় নেকি?	তাই বৰ ধুনীয়া নেকি? *"Is she very beautiful?"*	Adjectival	Affirmative	Deleted (Zero = 1)
65	**Informal Dialogue** *Tag: Interrogative Adjectival*	বতৰটো গৰম হয় নেকি?	বতৰটো গৰম নেকি? *"Is it sunny today?"*	Adjectival	Affirmative	Deleted (Zero = 1)
66	**Informal Dialogue** *Tag: Nominal/identifty*	মই এই ঘৰৰ নতুন ভাড়াতীয়া হওঁ।	মই এই ঘৰৰ নতুন ভাড়াতীয়া। *"I am the new tenant of this house."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
67	**Informal Dialogue** *Tag: Nominal/kinship*	তুমি মোৰ সৰু ভনী হোৱা।	তুমি মোৰ সৰু ভনী। *"You are my younger sister"*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
68	**Informal Dialogue** *Tag: Nominal/proffesion*	এইজন আমাৰ গাঁৱৰ ডাক্তৰ হয়।	এইজন আমাৰ গাঁৱৰ ডাক্তৰ। *"This person is the doctor of our village"*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
69	**Informal Dialogue** *Tag: Nominal/role*	ৰিমা আমাৰ দলৰ নেতা হয়।	ৰিমা আমাৰ দলৰ নেতা। *"Rima is the leader of our group."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
70	**Informal Dialogue** *Tag: Nominal/choice*	এইটো তোৰ নিজৰ সিদ্ধান্ত হয়।	এইটো তোৰ নিজৰ সিদ্ধান্ত। *"This is your own decision."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
71	**Informal Dialogue** *Tag: Nominal/role*	তাই আমাৰ ক্লাছৰ মনিটৰ হয়।	তাই আমাৰ ক্লাছৰ মনিটৰ। *"She is the monitor of our class."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
72	**Informal Dialogue** *Tag: Nominal/possesion*	সেইখন আমাৰ পুৰণা গাড়ী হয়।	সেইখন আমাৰ পুৰণা গাড়ী। *"That is our old car."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
73	**Informal Dialogue** *Tag: Nominal/role*	মই এই বিষয়টোৰ দায়িত্বত থকা মানুহ হওঁ।	মই এই বিষয়টোৰ দায়িত্বত থকা মানুহ। *"I am the person responsible for this matter."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
74	**Informal Dialogue** *Tag: Nominal/social relation*	তেওঁ আমাৰ ওচৰৰ মানুহ হয়।	তেওঁ আমাৰ ওচৰৰ মানুহ। *"He/she is someone from our neighbourhood."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
75	**Informal Dialogue** *Tag: Nominal/evaluation*	এইটো মোৰ বাবে ভাল সুযোগ হয়।	এইটো মোৰ বাবে ভাল সুযোগ। *"This is a good opportunity for me."*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
76	**Informal Dialogue** *Tag: Adjectival/weather*	আজি আকাশখন একেবাৰে ডাৰৰীয়া হয়।	আজি আকাশখন একেবাৰে ডাৰৰীয়া। *"The sky is completely cloudy today."*	Adjectival	Affirmative	Deleted (Zero = 1)
77	**Informal Dialogue** *Tag: Adjectival/evaluation*	তোৰ কথাটো অলপ অদ্ভুত হয়।	তোৰ কথাটো অলপ অদ্ভুত। *"What you said is a little strange."*	Adjectival	Affirmative	Deleted (Zero = 1)
78	**Informal Dialogue** *Tag: Adjectival/texture*	এই কাপোৰযোৰ বৰ কোমল হয়।	এই কাপোৰযোৰ বৰ কোমল। *"This set of clothes is very soft."*	Adjectival	Affirmative	Deleted (Zero = 1)
79	**Informal Dialogue** *Tag: Adjectival/condition*	ৰাস্তাটো আজি পিছল হয়।	ৰাস্তাটো আজি পিছল। *"The road is slippery today."*	Adjectival	Affirmative	Deleted (Zero = 1)

#	H1: Register	Overt Copula Form (Assamese)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status
80	**Informal Dialogue** *Tag: Adjectival/behaviour*	তাৰ ব্যৱহাৰ বৰ ভাল হয়।	তাৰ ব্যৱহাৰ বৰ ভাল। *"His behaviour is very good."*	Adjectival	Affirmative	Deleted (Zero = 1)
81	**Informal Dialogue** *Tag: Adjectival/taste*	এই মিঠাইটো বৰ মিঠা হয়।	এই মিঠাইটো বৰ মিঠা। *"This sweet is very sweet."*	Adjectival	Affirmative	Deleted (Zero = 1)
82	**Informal Dialogue** *Tag: Adjectival/quality*	তাইৰ মাতটো অলপ ডাঙৰ হয়।	তাইৰ মাতটো অলপ ডাঙৰ। *"Her voice is a little loud."*	Adjectival	Affirmative	Deleted (Zero = 1)
83	**Informal Dialogue** *Tag: Adjectival/evaluation*	এই সমস্যাটো বৰ জৰুৰী হয়।	এই সমস্যাটো বৰ জৰুৰী। *"This problem is very urgent."*	Adjectival	Affirmative	Deleted (Zero = 1)
84	**Informal Dialogue** *Tag: Locative/existence*	ছাত্রকেইজন ক্লাছৰ বাহিৰত আছে।	ছাত্রকেইজন ক্লাছৰ বাহিৰত। *"The students are outside the classroom."*	Locative/Existential	Affirmative	Deleted (Zero = 1)
85	**Informal Dialogue** *Tag: Locative/possesion*	তোৰ চাবিপাত মোৰ হাতত আছে।	তোৰ চাবিপাত মোৰ হাতত। *"Your key is with me."*	Locative/Existential	Affirmative	Deleted (Zero = 1)
86	**Informal Dialogue** *Tag: Interrogative/nominal*	এইজনেই তোমাৰ নতুন বন্ধু হয় নেকি?	এইজনেই তোমাৰ নতুন বন্ধু নেকি? *"Is this your new friend?"*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
87	**Informal Dialogue** *Tag: Interrogative/nominal*	এইটো তোৰ শেষ সিদ্ধান্ত হয় নেকি?	এইটো তোৰ শেষ সিদ্ধান্ত নেকি? *"Is this your final decision?"*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
88	**Informal Dialogue** *Tag: Interrogative/adjectival*	এইটো খাবলৈ ভাল হয় নেকি?	এইটো খাবলৈ ভাল নেকি? *"Is this good to eat?"*	Adjectival	Affirmative	Deleted (Zero = 1)
89	**Informal Dialogue** *Tag: Interrogative/adjectival*	আজি ৰাস্তাটো খালী হয় নেকি?	আজি ৰাস্তাটো খালী নেকি? *"Is the road empty today?"*	Adjectival	Affirmative	Deleted (Zero = 1)
90	**Informal Dialogue** *Tag: Interrogative/nominal*	সেইজন তোমালোকৰ নতুন মেনেজাৰ হয় নেকি?	সেইজন তোমালোকৰ নতুন মেনেজাৰ নেকি? *"Is that person your new manager?"*	Equational/Nominal	Affirmative	Deleted (Zero = 1)
91	**Informal Dialogue** *Tag: Interrogative/state*	তোৰ ফোনটো এতিয়া অন আছে নেকি?	তোৰ ফোনটো এতিয়া অন নেকি? *"Is your phone switched on now?"*	Locative/Existential	Affirmative	Deleted (Zero = 1)
92	**Informal Dialogue** *Tag: Interrogative/locative*	তেওঁ এতিয়া ঘৰত আছেনে?	তেওঁ এতিয়া ঘৰত? *"Is he/she at home now?"*	Locative/Existential	Affirmative	Deleted (Zero = 1)
93	**Informal Dialogue** *Tag: Negative locative (নাই)*	তেওঁ আজি অফিচত নাই।	——- *"He/she is not at the office today."*	Locative/Existential	Negative	Retained (Overt = 0)
94	**Informal Dialogue** *Tag: Negative adjectival (নহয়)*	এইটো একেবাৰে সহজ নহয়।	——- *"This is not easy at all."*	Adjectival	Negative	Retained (Overt = 0)
95	**Informal Dialogue** *Tag: Negative nominal (নহয়)*	তাই আমাৰ ক্লাছৰ মনিটৰ নহয়।	——- *"She is not the monitor of our class."*	Equational/Nominal	Negative	Retained (Overt = 0)
96	**Informal Dialogue** *Tag: Negative possessive (নাই)*	মোৰ ওচৰত এতিয়া নগদ টকা নাই।	——- *"I don't have cash with me now."*	Locative/Existential	Negative	Retained (Overt = 0)

#	H1: Register	Overt Copula Form (Assamese)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status
97	**Informal Dialogue** *Tag: Negative existential (নাই)*	ইয়াত কোনো সমস্যা নাই।	——- *"There is no problem here."*	Locative/Existential	Negative	Retained (Overt = 0)

A.4 Bengali Audited Present-Tense Non-Verbal Sentences (Representative Stratified Sample: n = 64 across all 8 Sources)

#	Source & Author	H1: Sub-Register	Bengali Sentence & Target Clause	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Form
1	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0102)*	**Literary Dialogue**	এরকম করাটা বিপজ্জনক। *Clause: এরকম করাটা বিপজ্জনক।*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: ZERO*
2	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0085)*	**Literary Narrative**	আমার ওজন এত কম *Clause: আমার ওজন এত কম*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: ZERO*
3	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0091)*	**Literary Narrative**	খুব রোগা নয় *Clause: খুব রোগা নয়*	Adjectival	Negative	Retained (Overt = 0) *Form: নয়*
4	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0122)*	**Literary Dialogue**	আসল কথা হলো ওই পারসপেকটিভ *Clause: আসল কথা হলো ওই পারসপেকটিভ*	Equational/Nominal	Affirmative	Retained (Overt = 0) *Form: হলো*
5	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0111)*	**Literary Dialogue**	আপনি তো অরবিন্দবাবু। *Clause: আপনি তো অরবিন্দবাবু।*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: ZERO*
6	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0084)*	**Literary Narrative**	আমি বেঁটে মানুষ *Clause: আমি বেঁটে মানুষ*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: ZERO*
7	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0112)*	**Literary Narrative**	মোটেই তা নয়। *Clause: মোটেই তা নয়।*	Equational/Nominal	Negative	Retained (Overt = 0) *Form: নয়*
8	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0099)*	**Literary Narrative**	গোলকধাঁধার মতো গলিগুলো আছে *Clause: গোলকধাঁধার মতো গলিগুলো আছে*	Locative/Existential	Affirmative	Retained (Overt = 0) *Form: আছে*
9	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0082)*	**Literary Narrative**	এই যে আমি এখানে। *Clause: এই যে আমি এখানে।*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Form: ZERO*
10	**আমাকে দেখুন** *শীর্ষেন্দু মুখোপাধ্যায় (BN_LIT_0089)*	**Literary Narrative**	আমার চেহারায় এমন কিছু নেই *Clause: আমার চেহারায় এমন কিছু নেই*	Locative/Existential	Negative	Retained (Overt = 0) *Form: নেই*
11	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী (BN_LIT_0066)*	**Literary Dialogue**	এটা খুব ভালো। *Clause: এটা খুব ভালো।*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: ZERO*
12	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী (BN_LIT_0068)*	**Literary Narrative**	দুপুরবেলা, সবাই যখন নিথর নিথর *Clause: দুপুরবেলা, সবাই যখন নিথর নিথর*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: ZERO*
13	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী (BN_LIT_0074)*	**Literary Dialogue**	ওরকম একটি লেখক মেসো থাকা মন্দ নয়। *Clause: ওরকম একটি লেখক মেসো থাকা মন্দ নয়।*	Adjectival	Negative	Retained (Overt = 0) *Form: নয়*
14	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী (BN_LIT_0076)*	**Literary Dialogue**	লজ্জা কী? *Clause: লজ্জা কী?*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: ZERO*
15	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী (BN_LIT_0053)*	**Literary Narrative**	তপনের নতুন মেসোমশাই একজন লেখক। *Clause: তপনের নতুন মেসোমশাই একজন লেখক।*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: ZERO*
16	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী (BN_LIT_0059)*	**Literary Narrative**	‘লেখক’ মানে কোনো আকাশ থেকে পড়া জীব নয় *Clause: ‘লেখক’ মানে কোনো আকাশ থেকে পড়া জীব নয়*	Equational/Nominal	Negative	Retained (Overt = 0) *Form: নয়*
17	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী (BN_LIT_0064)*	**Literary Dialogue**	তপনের হাত আছে। *Clause: তপনের হাত আছে।*	Locative/Existential	Affirmative	Retained (Overt = 0) *Form: আছে*

#	Source & Author	H1: Sub-Register	Bengali Sentence & Target Clause	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Form
18	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী* *(BN_LIT_0058)*	**Literary Narrative**	ছুটি আছে *Clause: ছুটি আছে*	Locative/Existential	Affirmative	Retained (Overt = 0) *Form: **আছে***
19	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী* *(BN_LIT_0062)*	**Literary Narrative**	রত্নের মূল্য জহুরির কাছেই। *Clause: রত্নের মূল্য জহুরির কাছেই।*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Form: ZERO*
20	**জ্ঞানচক্ষু** *আশাপূর্ণা দেবী* *(BN_LIT_0056)*	**Literary Narrative**	কোথাও কিছু উলটোপালটা নেই *Clause: কোথাও কিছু উলটোপালটা নেই*	Locative/Existential	Negative	Retained (Overt = 0) *Form: **নেই***
21	**নদীর বিদ্রোহ** *মানিক বন্দ্যোপাধ্যায়* *(BN_LIT_0265)*	**Literary Narrative**	ত্রিশ বছর বয়সে নদীর জন্য নদেরচাঁদের এত বেশি মায়া একটু অস্বাভাবিক। *Clause: ত্রিশ বছর বয়সে নদীর জন্য নদেরচাঁদের এত বেশি মায়া একটু অস্বাভাবিক।*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: ZERO*
22	**নদীর বিদ্রোহ** *মানিক বন্দ্যোপাধ্যায়* *(BN_LIT_0266)*	**Literary Narrative**	সে তো একটা স্টেশনের স্টেশন মাস্টার *Clause: সে তো একটা স্টেশনের স্টেশন মাস্টার*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: ZERO*
23	**নদীর বিদ্রোহ** *মানিক বন্দ্যোপাধ্যায়* *(BN_LIT_0271)*	**Literary Narrative**	তাহাকে বিশ্বাস নাই। *Clause: তাহাকে বিশ্বাস নাই।*	Locative/Existential	Negative	Retained (Overt = 0) *Form: **নাই***
24	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায়* *(BN_LIT_0190)*	**Literary Dialogue**	শখ ষোলোআনাই বজায় আছে *Clause: শখ ষোলোআনাই বজায় আছে*	Adjectival	Affirmative	Retained (Overt = 0) *Form: **আছে***
25	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায়* *(BN_LIT_0197)*	**Literary Dialogue**	গাঁজা খাবার সমস্ত লক্ষণই তোমাতে বিদ্যমান বাবা *Clause: গাঁজা খাবার সমস্ত লক্ষণই তোমাতে বিদ্যমান বাবা*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: ZERO*
26	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায়* *(BN_LIT_0188)*	**Literary Narrative**	এ-সকল বিবরণ দিতে যাওয়াই বৃথা *Clause: এ-সকল বিবরণ দিতে যাওয়াই বৃথা*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: ZERO*
27	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায়* *(BN_LIT_0210)*	**Literary Dialogue**	আমার দেশের চেয়ে তো তিনি আপনার নন। *Clause: আমার দেশের চেয়ে তো তিনি আপনার নন।*	Adjectival	Negative	Retained (Overt = 0) *Form: **নন***
28	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায়* *(BN_LIT_0208)*	**Literary Narrative**	এই ধারণা যে সত্য নয় *Clause: এই ধারণা যে সত্য নয়*	Adjectival	Negative	Retained (Overt = 0) *Form: **নয়***
29	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায়* *(BN_LIT_0193)*	**Literary Dialogue**	তোমার নাম কী হে? *Clause: তোমার নাম কী হে?*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: ZERO*
30	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায়* *(BN_LIT_0192)*	**Literary Dialogue**	সে যে এ নয় *Clause: সে যে এ নয়*	Equational/Nominal	Negative	Retained (Overt = 0) *Form: **নয়***
31	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায়* *(BN_LIT_0194)*	**Literary Dialogue**	তোমার ট্যাঁকে এবং পকেটে কী আছে? *Clause: তোমার ট্যাঁকে এবং পকেটে কী আছে?*	Locative/Existential	Affirmative	Retained (Overt = 0) *Form: **আছে***
32	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায়* *(BN_LIT_0195)*	**Literary Dialogue**	এ বস্তুটি পকেটে কেন? *Clause: এ বস্তুটি পকেটে কেন?*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Form: ZERO*

#	Source & Author	H1: Sub-Register	Bengali Sentence & Target Clause	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Form
33	**পথের দাবী (পাঠ্যাংশ)** *শরৎচন্দ্র চট্টোপাধ্যায় (BN_LIT_0199)*	**Literary Dialogue**	এই জানোয়ারটাকে ওয়াচ করবার দরকার নেই বড়োবাবু। *Clause: এই জানোয়ারটাকে ওয়াচ করবার দরকার নেই বড়োবাবু।*	Locative/Existential	Negative	Retained (Overt = 0) *Form: **নেই***
34	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0174)*	**Literary Dialogue**	এই যথেষ্ট। *Clause: এই যথেষ্ট।*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
35	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0150)*	**Literary Narrative**	চোখ দুটো কটকটে লাল। *Clause: চোখ দুটো কটকটে লাল।*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
36	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0172)*	**Literary Dialogue**	হরিদার গলার স্বর এরকমেরই নয়। *Clause: হরিদার গলার স্বর এরকমেরই নয়।*	Adjectival	Negative	Retained (Overt = 0) *Form: **নয়***
37	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0146)*	**Literary Narrative**	রোজই একটা চাকরির কাজ করে যাওয়া হরিদার পক্ষে সম্ভব নয়। *Clause: রোজই একটা চাকরির কাজ করে যাওয়া হরিদার পক্ষে সম্ভব নয়।*	Adjectival	Negative	Retained (Overt = 0) *Form: **নয়***
38	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0178)*	**Literary Dialogue**	ওসব হলো সুন্দর সুন্দর এক-একটি বঞ্চনা। *Clause: ওসব হলো সুন্দর সুন্দর এক-একটি বঞ্চনা।*	Equational/Nominal	Affirmative	Retained (Overt = 0) *Form: **হলো***
39	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0137)*	**Literary Dialogue**	সন্ন্যাসীর বয়সও হাজার বছরের বেশি *Clause: সন্ন্যাসীর বয়সও হাজার বছরের বেশি*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
40	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0165)*	**Literary Dialogue**	আমি মহারাজ নই *Clause: আমি মহারাজ নই*	Equational/Nominal	Negative	Retained (Overt = 0) *Form: **নই***
41	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0138)*	**Literary Dialogue**	সন্ন্যাসী কি এখনও আছেন? *Clause: সন্ন্যাসী কি এখনও আছেন?*	Locative/Existential	Affirmative	Retained (Overt = 0) *Form: **আছেন***
42	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0181)*	**Literary Dialogue**	আমার বুকের ভিতরেই যে সব তীর্থ। *Clause: আমার বুকের ভিতরেই যে সব তীর্থ।*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
43	**বহুরূপী** *সুবোধ ঘোষ (BN_LIT_0168)*	**Literary Dialogue**	রাগ নামে কোনো রিপু আমার নেই। *Clause: রাগ নামে কোনো রিপু আমার নেই।*	Locative/Existential	Negative	Retained (Overt = 0) *Form: **নেই***
44	**বাংলা ভাষায় বিজ্ঞান** *রাজশেখর বসু (BN_LIT_0235)*	**Literary Essay**	পিতলের চাইতে আলিউমিনিয়াম হালকা *Clause: পিতলের চাইতে আলিউমিনিয়াম হালকা*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
45	**বাংলা ভাষায় বিজ্ঞান** *রাজশেখর বসু (BN_LIT_0237)*	**Literary Essay**	বাংলায় বিজ্ঞান শেখা তাদের সংস্কারের বিরোধী নয়। *Clause: বাংলায় বিজ্ঞান শেখা তাদের সংস্কারের বিরোধী নয়।*	Adjectival	Negative	Retained (Overt = 0) *Form: **নয়***
46	**বাংলা ভাষায় বিজ্ঞান** *রাজশেখর বসু (BN_LIT_0258)*	**Literary Essay**	পরিভাষার উদ্দেশ্য ভাষার সংক্ষেপ এবং অর্থ সুনির্দিষ্ট করা। *Clause: পরিভাষার উদ্দেশ্য ভাষার সংক্ষেপ এবং অর্থ সুনির্দিষ্ট করা।*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
47	**বাংলা ভাষায় বিজ্ঞান** *রাজশেখর বসু (BN_LIT_0245)*	**Literary Essay**	পরিভাষা রচনা একজনের কাজ নয় *Clause: পরিভাষা রচনা একজনের কাজ নয়*	Equational/Nominal	Negative	Retained (Overt = 0) *Form: **নয়***
48	**বাংলা ভাষায় বিজ্ঞান** *রাজশেখর বসু (BN_LIT_0242)*	**Literary Essay**	বাংলা ভাষায় বিজ্ঞানচর্চায় এখনও নানা রকম বাধা আছে। *Clause: বাংলা ভাষায় বিজ্ঞানচর্চায় এখনও নানা রকম বাধা আছে।*	Locative/Existential	Affirmative	Retained (Overt = 0) *Form: **আছে***

#	Source & Author	H1: Sub-Register	Bengali Sentence & Target Clause	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Form
49	**বাংলা ভাষায় বিজ্ঞান** *রাজশেখর বসু* *(BN_LIT_0234)*	**Literary Essay**	প্রথম শ্রেণির পাঠকদের বিজ্ঞানের সঙ্গে পূর্ব পরিচয় নেই। *Clause: প্রথম শ্রেণির পাঠকদের বিজ্ঞানের সঙ্গে পূর্ব পরিচয় নেই।*	Locative/Existential	Negative	Retained (Overt = 0) *Form: **নেই***
50	**শাবলতলার মাঠ** *বিভূতিভূষণ বন্দ্যোপাধ্যায়* *(BN_LIT_0047)*	**Literary Dialogue**	জায়গাটা বড়ো চমৎকার *Clause: জায়গাটা বড়ো চমৎকার*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
51	**শাবলতলার মাঠ** *বিভূতিভূষণ বন্দ্যোপাধ্যায়* *(BN_LIT_0013)*	**Literary Dialogue**	এ হলো প্রহসন। *Clause: এ হলো প্রহসন।*	Equational/Nominal	Affirmative	Retained (Overt = 0) *Form: **হলো***
52	**শাবলতলার মাঠ** *বিভূতিভূষণ বন্দ্যোপাধ্যায়* *(BN_LIT_0004)*	**Literary Dialogue**	এটা শাবলতলার মাঠ, না? *Clause: এটা শাবলতলার মাঠ, না?*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
53	**শাবলতলার মাঠ** *বিভূতিভূষণ বন্দ্যোপাধ্যায়* *(BN_LIT_0037)*	**Literary Dialogue**	দু ঘণ্টার ওপর তো এখানে আছি। *Clause: দু ঘণ্টার ওপর তো এখানে আছি।*	Locative/Existential	Affirmative	Retained (Overt = 0) *Form: **আছি***
54	**শাবলতলার মাঠ** *বিভূতিভূষণ বন্দ্যোপাধ্যায়* *(BN_LIT_0005)*	**Literary Dialogue**	গ্রামগুলো সব কোথায়? *Clause: গ্রামগুলো সব কোথায়?*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
55	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray* *(BN_SUB_0003)*	**Film Dialogue**	এটা তো অপয়া। *Clause: এটা তো অপয়া*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
56	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray* *(BN_SUB_0046)*	**Film Song**	উঁচা নিচা ছোট বড় সমান। *Clause: উঁচা নিচা ছোট বড় সমান*	Adjectival	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
57	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray* *(BN_SUB_0031)*	**Film Dialogue**	যথেষ্ট নয়। *Clause: যথেষ্ট নয়*	Adjectival	Negative	Retained (Overt = 0) *Form: **নয়***
58	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray* *(BN_SUB_0083)*	**Film Song**	সেও সুখী নয়। *Clause: সেও সুখী নয়*	Adjectival	Negative	Retained (Overt = 0) *Form: **নয়***
59	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray* *(BN_SUB_0070)*	**Film Dialogue**	আমরা হলাম যে গুপ্তচর। *Clause: আমরা হলাম যে গুপ্তচর*	Equational/Nominal	Affirmative	Retained (Overt = 0) *Form: **হলাম***
60	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray* *(BN_SUB_0001)*	**Film Dialogue**	তুমি চাষা। *Clause: তুমি চাষা*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
61	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray* *(BN_SUB_0051)*	**Film Dialogue**	তা নয়। *Clause: তা নয়*	Equational/Nominal	Negative	Retained (Overt = 0) *Form: **নয়***

#	Source & Author	H1: Sub-Register	Bengali Sentence & Target Clause	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Form
62	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray (BN_SUB_0005)*	**Film Dialogue**	সকাল কি এখন আছে। *Clause: সকাল কি এখন আছে*	Locative/Existential	Affirmative	Retained (Overt = 0) *Form: **আছে***
63	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray (BN_SUB_0011)*	**Film Dialogue**	আমার তানপুরা কই। *Clause: আমার তানপুরা কই*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Form: **ZERO***
64	**Goopy Gyne Bagha Byne (1969)** *Satyajit Ray (BN_SUB_0008)*	**Film Dialogue**	সুর নেই। *Clause: সুর নেই*	Locative/Existential	Negative	Retained (Overt = 0) *Form: **নেই***

A.5 Complete Telugu Manually Verified Dataset (All N = 116 Non-Verbal Clauses)

#	Source & Manual ID	H1: Register & Sub-Register	Telugu Sentence (Telugu Script)	H2: Predicate Type	H3: Polarity	Assigned Copula Status
1	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0001*	**Film Dialogue** *Spoken Subtitle*	మీ నాన్న వెనకాలలో ఉన్న హోటల్ లోనే ఉన్నాడు.	Locative/Existential	Affirmative	Retained (Overt = 0)
2	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0002*	**Film Dialogue** *Spoken Subtitle*	వాల్మీకి మీ నాన్న కాదు	Equational/Nominal	Negative	Retained (Overt = 0)
3	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0003*	**Film Dialogue** *Spoken Subtitle*	వీడు ఉత్తమవుడు కాదు	Equational/Nominal	Negative	Retained (Overt = 0)
4	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0004*	**Film Dialogue** *Spoken Subtitle*	ఎవడా గ్రే కోటోడా	Equational/Nominal	Affirmative	Deleted (Zero = 1)
5	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0005*	**Film Dialogue** *Spoken Subtitle*	పాయింటే అబద్ధం సార్	Equational/Nominal	Affirmative	Deleted (Zero = 1)
6	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0006*	**Film Dialogue** *Spoken Subtitle*	నాకు ఇంతకన్నా ఏం అవసరం లేదు	Locative/Existential	Negative	Retained (Overt = 0)
7	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0007*	**Film Dialogue** *Spoken Subtitle*	లోపలికి ఎంట్రీ లేదు	Locative/Existential	Negative	Retained (Overt = 0)
8	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0008*	**Film Dialogue** *Spoken Subtitle*	నా బాధకంతు లేదు	Locative/Existential	Negative	Retained (Overt = 0)
9	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0009*	**Film Dialogue** *Spoken Subtitle*	ఇది వీడి దురదృష్టం.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
10	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0010*	**Film Dialogue** *Spoken Subtitle*	ఇదో లెక్క రేయ్	Equational/Nominal	Affirmative	Deleted (Zero = 1)
11	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0011*	**Film Dialogue** *Spoken Subtitle*	ఏం కాంట్రాస్ట్ గా ఉన్నారు	Adjectival	Affirmative	Retained (Overt = 0)
12	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0012*	**Film Dialogue** *Spoken Subtitle*	వాడు ఆల్రెడీ డౌన్ లో ఉన్నాడు	Locative/Existential	Affirmative	Retained (Overt = 0)
13	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0013*	**Film Dialogue** *Spoken Subtitle*	నిజంగానే తేలిగ్గా ఉంది.	Adjectival	Affirmative	Retained (Overt = 0)
14	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0014*	**Film Dialogue** *Spoken Subtitle*	చెత్త నాలో ఉందా	Locative/Existential	Affirmative	Retained (Overt = 0)

#	Source & Manual ID	H1: Register & Sub-Register	Telugu Sentence (Telugu Script)	H2: Predicate Type	H3: Polarity	Assigned Copula Status
15	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0015*	**Film Dialogue** *Spoken Subtitle*	మా నాన్న అంటే నీకు జెలసీ	Equational/Nominal	Affirmative	Deleted (Zero = 1)
16	**Ala_Vaikunthapurramuloo** *TE_MANUAL_0016*	**Film Dialogue** *Spoken Subtitle*	నాలో ఏం లేదు	Locative/Existential	Negative	Retained (Overt = 0)
17	**Srirastu_Subhamastu** *TE_MANUAL_0017*	**Film Dialogue** *Spoken Subtitle*	నువ్వు నాన్న అంటే నాకు ఇష్టం లేదా?	Locative/Existential	Negative	Retained (Overt = 0)
18	**Srirastu_Subhamastu** *TE_MANUAL_0018*	**Film Dialogue** *Spoken Subtitle*	ఏమన్నా ఆ లాస్ట్ సెమిస్టర్ కదా	Equational/Nominal	Affirmative	Deleted (Zero = 1)
19	**Srirastu_Subhamastu** *TE_MANUAL_0019*	**Film Dialogue** *Spoken Subtitle*	సిగ్నల్ సరిగ్గా లేవు	Adjectival	Negative	Retained (Overt = 0)
20	**Srirastu_Subhamastu** *TE_MANUAL_0020*	**Film Dialogue** *Spoken Subtitle*	నా ఫ్రెండ్ అలాంటిది కాదు.	Adjectival	Negative	Retained (Overt = 0)
21	**Srirastu_Subhamastu** *TE_MANUAL_0021*	**Film Dialogue** *Spoken Subtitle*	అమ్మ ఎలా ఉందిరా	Adjectival	Affirmative	Retained (Overt = 0)
22	**Srirastu_Subhamastu** *TE_MANUAL_0022*	**Film Dialogue** *Spoken Subtitle*	అబ్బబ్బ మీ ఇష్టం సార్	Equational/Nominal	Affirmative	Deleted (Zero = 1)
23	**Srirastu_Subhamastu** *TE_MANUAL_0023*	**Film Dialogue** *Spoken Subtitle*	చాలా ఎకసకాలు ఉన్నారా	Locative/Existential	Affirmative	Retained (Overt = 0)
24	**Srirastu_Subhamastu** *TE_MANUAL_0024*	**Film Dialogue** *Spoken Subtitle*	నీకు అది ముఖ్యమా నందు ముఖ్యమా	Adjectival	Affirmative	Deleted (Zero = 1)
25	**Srirastu_Subhamastu** *TE_MANUAL_0025*	**Film Dialogue** *Spoken Subtitle*	అవట్ హౌస్ లో వాచ్మెన్ రూమ్ ఉంది	Locative/Existential	Affirmative	Retained (Overt = 0)
26	**Srirastu_Subhamastu** *TE_MANUAL_0026*	**Film Dialogue** *Spoken Subtitle*	వీడు పెద్ద సైకో లో ఉన్నాడు	Adjectival	Affirmative	Retained (Overt = 0)
27	**Srirastu_Subhamastu** *TE_MANUAL_0027*	**Film Dialogue** *Spoken Subtitle*	మీకు వేరే దారి లేదు	Locative/Existential	Negative	Retained (Overt = 0)
28	**Srirastu_Subhamastu** *TE_MANUAL_0028*	**Film Dialogue** *Spoken Subtitle*	శ్యాము వీడు మా బామర్ది సుభాష్ అని ఎన్కౌంటర్ స్పెషలిస్ట్ అరుణ్ కి మేనమామ	Equational/Nominal	Affirmative	Deleted (Zero = 1)

#	Source & Manual ID	H1: Register & Sub-Register	Telugu Sentence (Telugu Script)	H2: Predicate Type	H3: Polarity	Assigned Copula Status
29	**Srirastu_Subhamastu** *TE_MANUAL_0029*	**Film Dialogue** *Spoken Subtitle*	అండ్ సెలబ్రేషన్స్ మా నువ్వు చదువు తప్పితే వేరే గ్యాస్ లేదు	Locative/Existential	Negative	Retained (Overt = 0)
30	**Srirastu_Subhamastu** *TE_MANUAL_0030*	**Film Dialogue** *Spoken Subtitle*	సుందర్ ఫిలాంత్రఫగడి కాదు	Equational/Nominal	Negative	Retained (Overt = 0)
31	**Srirastu_Subhamastu** *TE_MANUAL_0031*	**Film Dialogue** *Spoken Subtitle*	మాకు మాత్రం ఎవరున్నారురా	Locative/Existential	Affirmative	Retained (Overt = 0)
32	**Srirastu_Subhamastu** *TE_MANUAL_0032*	**Film Dialogue** *Spoken Subtitle*	వాడికి నాకు ఏ సంబంధం లేదు	Locative/Existential	Negative	Retained (Overt = 0)
33	**Srirastu_Subhamastu** *TE_MANUAL_0033*	**Film Dialogue** *Spoken Subtitle*	నీ భార్యగా అమ్మాయిని ఇంటికి తెచ్చుకునే హక్కు నీకు ఉంది.	Locative/Existential	Affirmative	Retained (Overt = 0)
34	**Srirastu_Subhamastu** *TE_MANUAL_0034*	**Film Dialogue** *Spoken Subtitle*	అంకుల్ అను ఈ పెళ్లి విషయంలో చాలా అప్సెట్ గా ఉంది.	Adjectival	Affirmative	Retained (Overt = 0)
35	**Srirastu_Subhamastu** *TE_MANUAL_0035*	**Film Dialogue** *Spoken Subtitle*	ఆ మొత్తం పాజిటివ్ బాబు	Adjectival	Affirmative	Deleted (Zero = 1)
36	**Vivaha_Bhojanambu** *TE_MANUAL_0036*	**Film Dialogue** *Spoken Subtitle*	మామయ్యగారు నువ్వు ఉండవమ్మ ఇంట్లో చిన్న పిల్లలు ముసలాళ్ళు ఉన్నారు.	Locative/Existential	Affirmative	Retained (Overt = 0)
37	**Vivaha_Bhojanambu** *TE_MANUAL_0037*	**Film Dialogue** *Spoken Subtitle*	ఇక్కడ ₹ రూపాయలు అంట వేస్ట్ కదా	Adjectival	Affirmative	Deleted (Zero = 1)
38	**Vivaha_Bhojanambu** *TE_MANUAL_0038*	**Film Dialogue** *Spoken Subtitle*	నువ్వు నా జోడీ	Equational/Nominal	Affirmative	Deleted (Zero = 1)
39	**Vivaha_Bhojanambu** *TE_MANUAL_0039*	**Film Dialogue** *Spoken Subtitle*	మీరు ఇంటర్వ్యూ మొదలెడితే మూడున్నర గంటలు	Equational/Nominal	Affirmative	Deleted (Zero = 1)
40	**Vivaha_Bhojanambu** *TE_MANUAL_0040*	**Film Dialogue** *Spoken Subtitle*	ఫ్యామిలీ మొత్తంలో ముసలోడే బెటర్ మీ ఇంటి పేరు ఏమిటో	Adjectival	Affirmative	Deleted (Zero = 1)
41	**Vivaha_Bhojanambu** *TE_MANUAL_0041*	**Film Dialogue** *Spoken Subtitle*	నువ్వు నా పేమకమే కదా	Equational/Nominal	Affirmative	Deleted (Zero = 1)
42	**Vivaha_Bhojanambu** *TE_MANUAL_0042*	**Film Dialogue** *Spoken Subtitle*	అది ఆర్టిస్ట్ కాదు	Equational/Nominal	Negative	Retained (Overt = 0)

#	Source & Manual ID	H1: Register & Sub-Register	Telugu Sentence (Telugu Script)	H2: Predicate Type	H3: Polarity	Assigned Copula Status
43	**Vivaha_Bhojanambu** *TE_MANUAL_0043*	**Film Dialogue** *Spoken Subtitle*	ఏంటి మీ అడ్రెస్	Equational/Nominal	Affirmative	Deleted (Zero = 1)
44	**Vivaha_Bhojanambu** *TE_MANUAL_0044*	**Film Dialogue** *Spoken Subtitle*	హలో అరే మీకో గుడ్ న్యూస్ రా	Locative/Existential	Affirmative	Deleted (Zero = 1)
45	**Vivaha_Bhojanambu** *TE_MANUAL_0045*	**Film Dialogue** *Spoken Subtitle*	ఆలివ్ ఆయిల్ నీకేనా	Equational/Nominal	Affirmative	Deleted (Zero = 1)
46	**Vivaha_Bhojanambu** *TE_MANUAL_0046*	**Film Dialogue** *Spoken Subtitle*	నువ్వు ఇంత కంజూస్ ఏంట్రా	Adjectival	Affirmative	Deleted (Zero = 1)
47	**Vivaha_Bhojanambu** *TE_MANUAL_0047*	**Film Dialogue** *Spoken Subtitle*	తొందరగా వెళ్ళటమే మంచిది.	Adjectival	Affirmative	Deleted (Zero = 1)
48	**Vivaha_Bhojanambu** *TE_MANUAL_0048*	**Film Dialogue** *Spoken Subtitle*	ప్రేమ గురించి దోమ గురించి కాదు	Equational/Nominal	Negative	Retained (Overt = 0)
49	**Vivaha_Bhojanambu** *TE_MANUAL_0049*	**Film Dialogue** *Spoken Subtitle*	మీ అడ్రెస్ ఏంటి సార్	Equational/Nominal	Affirmative	Deleted (Zero = 1)
50	**Diddubatu** *TE_MANUAL_0050*	**Literary Narrative** *Literary*	అనుకొని, రెక్క మెల్లన తెరవ, నడవలో దీపము లేదు.	Locative/Existential	Negative	Retained (Overt = 0)
51	**Diddubatu** *TE_MANUAL_0051*	**Literary Dialogue** *Literary*	మీ అమ్మ కాదురా	Equational/Nominal	Negative	Retained (Overt = 0)
52	**Diddubatu** *TE_MANUAL_0052*	**Literary Narrative** *Literary*	గోపాలుడు సదయ హృదయుడు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
53	**Diddubatu** *TE_MANUAL_0053*	**Literary Dialogue** *Literary*	యేటో మాయలా వుంది బాబూ	Adjectival	Affirmative	Retained (Overt = 0)
54	**Diddubatu** *TE_MANUAL_0054*	**Literary Dialogue** *Literary*	భగవంతుడి సృష్టిలోకల్లా ఉత్కృష్టమైన వస్తువ విద్య నేర్చిన స్త్రీ రత్నమే.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
55	**Diddubatu** *TE_MANUAL_0055*	**Literary Dialogue** *Literary*	నువ్వు తాతల నాటి నొఖరువి.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
56	**Diddubatu** *TE_MANUAL_0056*	**Literary Dialogue** *Literary*	మరి పంతులు మనుసు మనుసులో లేదు.	Locative/Existential	Negative	Retained (Overt = 0)
57	**Mee_Peremiti** *TE_MANUAL_0057*	**Literary Dialogue** *Literary*	అదేం తర్జుమా కాదు గద?	Equational/Nominal	Negative	Retained (Overt = 0)
58	**Mee_Peremiti** *TE_MANUAL_0058*	**Literary Dialogue** *Literary*	ఆ రెండో పుస్తకమేమిటో?	Equational/Nominal	Affirmative	Deleted (Zero = 1)

#	Source & Manual ID	H1: Register & Sub-Register	Telugu Sentence (Telugu Script)	H2: Predicate Type	H3: Polarity	Assigned Copula Status
59	**Mee_Peremiti** *TE_MANUAL_0059*	**Literary Narrative** *Literary*	మా గురువు గారి వంటి గురువులు లోకంలో లేరు.	Locative/Existential	Negative	Retained (Overt = 0)
60	**Mee_Peremiti** *TE_MANUAL_0060*	**Literary Narrative** *Literary*	బిందువులుగా స్రవించిన కన్నీరు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
61	**Mee_Peremiti** *TE_MANUAL_0061*	**Literary Narrative** *Literary*	శాస్త్రం మాటకేం గాని, కావ్యాల్లో మంచి రసగ్రాహి.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
62	**Mee_Peremiti** *TE_MANUAL_0062*	**Literary Narrative** *Literary*	మరి సుగుణ సంపత్తికో అంటే, సత్యకాలపు మనిషి అన్నప్పుడు, ఆయనే సత్యకాలపు మనిషి.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
63	**Mee_Peremiti** *TE_MANUAL_0063*	**Literary Narrative** *Literary*	రామ్మూర్తికి భూములు లావుగా వున్నాయి.	Adjectival	Affirmative	Retained (Overt = 0)
64	**Mee_Peremiti** *TE_MANUAL_0064*	**Literary Narrative** *Literary*	అవి పువ్వులు కావు.	Equational/Nominal	Negative	Retained (Overt = 0)
65	**Mee_Peremiti** *TE_MANUAL_0065*	**Literary Narrative** *Literary*	అవి పళ్ళు కావు.	Equational/Nominal	Negative	Retained (Overt = 0)
66	**Mee_Peremiti** *TE_MANUAL_0066*	**Literary Narrative** *Literary*	వెంకయ్య స్కూలులో మేష్టరు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
67	**Mee_Peremiti** *TE_MANUAL_0067*	**Literary Narrative** *Literary*	మా స్కూలు జట్టులో ఆరేడుగురం యిక్కడనే వున్నాం.	Locative/Existential	Affirmative	Retained (Overt = 0)
68	**Mee_Peremiti** *TE_MANUAL_0068*	**Literary Narrative** *Literary*	ఇప్పటి మట్టుకు మీకు తెలియవలసిన దేమంటే- అక్కడి విష్ణుక్షేత్రం ఆధునికం.	Adjectival	Affirmative	Deleted (Zero = 1)
69	**Mee_Peremiti** *TE_MANUAL_0069*	**Literary Narrative** *Literary*	ఆ వూరి నల్ల కొండల నంతటను శిధిలమైన బౌద్ధ కట్టడములు కలవు.	Locative/Existential	Affirmative	Retained (Overt = 0)
70	**Mee_Peremiti** *TE_MANUAL_0070*	**Literary Narrative** *Literary*	ఈ దేశంలో పాండవులు వుండని గుహలూ, సీతామ్మవారు స్నానమాడని గుంటలూ లేవు.	Locative/Existential	Negative	Retained (Overt = 0)
71	**Mee_Peremiti** *TE_MANUAL_0071*	**Literary Narrative** *Literary*	నాటికీ నేటికీ మరి పికరు లేదు.	Locative/Existential	Negative	Retained (Overt = 0)
72	**Mee_Peremiti** *TE_MANUAL_0072*	**Literary Narrative** *Literary*	మంచి సాహిత్యం కద్దు.	Locative/Existential	Affirmative	Retained (Overt = 0)
73	**Mee_Peremiti** *TE_MANUAL_0073*	**Literary Dialogue** *Literary*	ఒక కథ వుంది	Locative/Existential	Affirmative	Retained (Overt = 0)
74	**Mee_Peremiti** *TE_MANUAL_0074*	**Literary Narrative** *Literary*	ఈ గ్రామంలో శైవ వైష్ణవ మతాలకు వైరం చిరకాలం నుంచి కద్దు.	Locative/Existential	Affirmative	Retained (Overt = 0)

#	Source & Manual ID	H1: Register & Sub-Register	Telugu Sentence (Telugu Script)	H2: Predicate Type	H3: Polarity	Assigned Copula Status
75	**Mee_Peremiti** *TE_MANUAL_0075*	**Literary Dialogue** *Literary*	అవును నీకంటూ తెలియని సంగతి లేదు.	Locative/Existential	Negative	Retained (Overt = 0)
76	**Mee_Peremiti** *TE_MANUAL_0076*	**Literary Narrative** *Literary*	రంగాచార్యులు గారు బహు యోగ్యులు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
77	**Mee_Peremiti** *TE_MANUAL_0077*	**Literary Narrative** *Literary*	ఆమెకు ఒక కొమార్తె.	Locative/Existential	Affirmative	Deleted (Zero = 1)
78	**Mee_Peremiti** *TE_MANUAL_0078*	**Literary Narrative** *Literary*	మాకూ వాళ్ళకూ రాకపోకలు గలవు.	Locative/Existential	Affirmative	Retained (Overt = 0)
79	**Mee_Peremiti** *TE_MANUAL_0079*	**Literary Narrative** *Literary*	విష్ణుస్థలం యొక్క స్థితి ఇది.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
80	**Mee_Peremiti** *TE_MANUAL_0080*	**Literary Narrative** *Literary*	శైవులలో వున్న ఐకమత్యం వైష్ణవుల్లో లేదు.	Locative/Existential	Negative	Retained (Overt = 0)
81	**Mee_Peremiti** *TE_MANUAL_0081*	**Literary Dialogue** *Literary*	యీ బ్రాహ్మలదీ, జ్ఞానము కాదు.	Equational/Nominal	Negative	Retained (Overt = 0)
82	**Mee_Peremiti** *TE_MANUAL_0082*	**Literary Dialogue** *Literary*	యీ అజ్ఞులా అవతార పురుషులు?	Equational/Nominal	Affirmative	Deleted (Zero = 1)
83	**Mee_Peremiti** *TE_MANUAL_0083*	**Literary Dialogue** *Literary*	అందులో సారధి నాయడు లక్షాధికారి.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
84	**Mee_Peremiti** *TE_MANUAL_0084*	**Literary Dialogue** *Literary*	అతని బావమరిది రామినాయడు గ్రామ మునసబు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
85	**Mee_Peremiti** *TE_MANUAL_0085*	**Literary Dialogue** *Literary*	కొంచఫూర్తూ, నిషా బాజీన్నీ - భోజన ప్రియుడు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
86	**Mee_Peremiti** *TE_MANUAL_0086*	**Literary Dialogue** *Literary*	యీ తాలూకాలో కల్లా పెద్ద సారా దుకాణం యీ వూళ్ళోనే వుంది.	Locative/Existential	Affirmative	Retained (Overt = 0)
87	**Mee_Peremiti** *TE_MANUAL_0087*	**Literary Narrative** *Literary*	నాటికీ నేటికీ, రెంటియందూ ప్రపత్తి యేకరీతిగానే వుంది.	Adjectival	Affirmative	Retained (Overt = 0)
88	**Mee_Peremiti** *TE_MANUAL_0088*	**Literary Dialogue** *Literary*	అది లేకుంటే వైష్ణవుడైనా, కార్యం లేదు	Locative/Existential	Negative	Retained (Overt = 0)
89	**Mee_Peremiti** *TE_MANUAL_0089*	**Literary Dialogue** *Literary*	రామానుజయ్య అయితే, చులాగ్గా డేగలా వున్నాడు గనక, అవస్యం అతగాడే గరుడాళ్వారి అవతారం.	Adjectival	Affirmative	Retained (Overt = 0)
90	**Mee_Peremiti** *TE_MANUAL_0090*	**Literary Dialogue** *Literary*	రామానుజయ్య తేలిగ్గా వున్నాడు	Adjectival	Affirmative	Retained (Overt = 0)
91	**Mee_Peremiti** *TE_MANUAL_0091*	**Literary Dialogue** *Literary*	యుక్తాయుక్తం యెరిగిన మనిషివి.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
92	**Mee_Peremiti** *TE_MANUAL_0092*	**Literary Dialogue** *Literary*	వైష్ణవ మత నిషిద్ధము.	Adjectival	Affirmative	Deleted (Zero = 1)

#	Source & Manual ID	H1: Register & Sub-Register	Telugu Sentence (Telugu Script)	H2: Predicate Type	H3: Polarity	Assigned Copula Status
93	**Mee_Peremiti** *TE_MANUAL_0093*	**Literary Narrative** *Literary*	ఆ మాట నిజమే!	Adjectival	Affirmative	Deleted (Zero = 1)
94	**Mee_Peremiti** *TE_MANUAL_0094*	**Literary Dialogue** *Literary*	నాకెందుకు తల్లీ, అపవిత్రుడను	Equational/Nominal	Affirmative	Deleted (Zero = 1)
95	**Mee_Peremiti** *TE_MANUAL_0095*	**Literary Dialogue** *Literary*	నీకు కొండంత గుండె వుంది!	Locative/Existential	Affirmative	Retained (Overt = 0)
96	**Mee_Peremiti** *TE_MANUAL_0096*	**Literary Narrative** *Literary*	లేకపోతాయని గదా నీ దుఃఖం.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
97	**Mee_Peremiti** *TE_MANUAL_0097*	**Literary Dialogue** *Literary*	అమ్మా శర్కరపొంగలి, దధ్యోదనమూ అనేవి ముఖ్యములు కావు.	Adjectival	Negative	Retained (Overt = 0)
98	**Mee_Peremiti** *TE_MANUAL_0098*	**Literary Narrative** *Literary*	దానిలో అరకాసంతయినా మహత్తు లేదా?	Locative/Existential	Negative	Retained (Overt = 0)
99	**Mee_Peremiti** *TE_MANUAL_0099*	**Literary Dialogue** *Literary*	గనక మీ వైష్ణవ మతం గొప్పా అతగాడి తురక మతం గొప్పా?	Adjectival	Affirmative	Deleted (Zero = 1)
100	**Mee_Peremiti** *TE_MANUAL_0100*	**Literary Dialogue** *Literary*	ఈ రెండు విధాల మనుష్యులూ శైవుల్లోనూ, వైష్ణవుల్లోనూ కూడా వున్నారు.	Locative/Existential	Affirmative	Retained (Overt = 0)
101	**Mee_Peremiti** *TE_MANUAL_0101*	**Literary Dialogue** *Literary*	అతగాడు కబీరు దాసంత భక్తుడు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
102	**Mee_Peremiti** *TE_MANUAL_0102*	**Literary Dialogue** *Literary*	అమ్మా మీరు సాక్షాత్తూ శ్రీ మహాలక్ష్మి అవతారం.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
103	**Mee_Peremiti** *TE_MANUAL_0103*	**Literary Dialogue** *Literary*	శివుడన్నా, విష్ణున్నా, పీరన్నా బుద్ధుడన్నా ఆ పరమాత్మ మట్టుకు ఒక్కడే గదా?	Equational/Nominal	Affirmative	Deleted (Zero = 1)
104	**Mee_Peremiti** *TE_MANUAL_0104*	**Literary Dialogue** *Literary*	పీరో, బిట్రో, దెయ్యమో, దేవరో, యీ అజ్ఞులకు మూఢభక్తి అయినా వుంది.	Locative/Existential	Affirmative	Retained (Overt = 0)
105	**Metilda** *TE_MANUAL_0105*	**Literary Narrative** *Literary*	రామారావు నాకు ప్రాణసమానమైన మిత్రుడు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
106	**Metilda** *TE_MANUAL_0106*	**Literary Narrative** *Literary*	అంత ముసలివాడు కాడు.	Equational/Nominal	Negative	Retained (Overt = 0)
107	**Metilda** *TE_MANUAL_0107*	**Literary Narrative** *Literary*	దానిలో మూడు బీరువాలతో పుస్తకాలు వున్నాయి.	Locative/Existential	Affirmative	Retained (Overt = 0)
108	**Metilda** *TE_MANUAL_0108*	**Literary Narrative** *Literary*	ఇది యింగ్లీషు పద్ధతి ప్రకారం యిద్దరికీ శరీర వ్యాయామం.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
109	**Metilda** *TE_MANUAL_0109*	**Literary Narrative** *Literary*	ఇంటిలో మిగిలిన వాళ్ళు ఒక ముసలి వంట బ్రాహ్మణుడు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)

#	Source & Manual ID	H1: Register & Sub-Register	Telugu Sentence (Telugu Script)	H2: Predicate Type	H3: Polarity	Assigned Copula Status
110	**Metilda** *TE_MANUAL_0110*	**Literary Dialogue** *Literary*	ఆలు మొగుళ్ళ దెబ్బలాటల్లోకి వెళ్లవద్దని మన పెద్దల శాసనం.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
111	**Metilda** *TE_MANUAL_0111*	**Literary Dialogue** *Literary*	అవి అభేద్యాలు, అగమ్య గోచరాలు.	Adjectival	Affirmative	Deleted (Zero = 1)
112	**Metilda** *TE_MANUAL_0112*	**Literary Dialogue** *Literary*	గుడ్డకి లోపం లేదు.	Locative/Existential	Negative	Retained (Overt = 0)
113	**Metilda** *TE_MANUAL_0113*	**Literary Dialogue** *Literary*	అది కూడని పని.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
114	**Metilda** *TE_MANUAL_0114*	**Literary Dialogue** *Literary*	ఆ అవకాశానికి అవకాశం యివ్వని వాడె ప్రాజ్ఞుడు	Equational/Nominal	Affirmative	Deleted (Zero = 1)
115	**Metilda** *TE_MANUAL_0115*	**Literary Dialogue** *Literary*	నా పెళ్ళాం బహు బుద్ధిమంతురాలు.	Equational/Nominal	Affirmative	Deleted (Zero = 1)
116	**Metilda** *TE_MANUAL_0116*	**Literary Dialogue** *Literary*	నాకు నీతో లోకం.	Equational/Nominal	Affirmative	Deleted (Zero = 1)

A.6 Complete Tamil Audited Dataset (All N = 106 Valid Non-Verbal Clauses)

#	H1: Sub-Register	Overt Copula Form (Tamil)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Verb
1	**Formal Prose**	மூடநம்பிக்கை ஒரு சமூக நோய் ஆகும்.	மூடநம்பிக்கை ஒரு சமூக நோய். *"Superstition is a social disease."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
2	**Formal Prose**	இவையே மூடநம்பிக்கைகள் ஆகும்.	இவையே மூடநம்பிக்கைகள். *"These are superstitions."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
3	**Formal Prose**	சீர்திருத்தம் ஒரு வாழ்க்கை முறை ஆகும்.	சீர்திருத்தம் ஒரு வாழ்க்கை முறை. *"Reform is a way of life."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
4	**Formal Prose**	பணிவு ஒரு மனப்பண்பு ஆகும்.	பணிவு ஒரு மனப்பண்பு. *"Humility is a quality of the mind."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
5	**Formal Prose**	பணிவு நல்லொழுக்கத்தின் அடையாளம் ஆகும்.	பணிவு நல்லொழுக்கத்தின் அடையாளம். *"Humility is a sign of good conduct."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
6	**Formal Prose**	காலம் விலைமதிப்பற்ற செல்வம் ஆகும்.	காலம் விலைமதிப்பற்ற செல்வம். *"Time is invaluable wealth."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
7	**Formal Prose**	கல்வியே ஒருவருக்கு அழியாச் செல்வம் ஆகும்.	கல்வியே ஒருவருக்கு அழியாச் செல்வம். *"Education alone is one's imperishable wealth."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
8	**Formal Prose**	நேர்மையே சிறந்த கொள்கை ஆகும்.	நேர்மையே சிறந்த கொள்கை. *"Honesty is the best policy."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
9	**Formal Prose**	மக்கள் சேவையே மகேசன் சேவை ஆகும்.	மக்கள் சேவையே மகேசன் சேவை. *"Service to people is service to God." — Zero form is the proverb as usually quoted*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
10	**Formal Prose**	உலகம் நம் அனைவருக்கும் பொதுவான வீடு ஆகும்.	உலகம் நம் அனைவருக்கும் பொதுவான வீடு. *"The world is the common home of us all."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
11	**Formal Prose**	இன்பமும் துன்பமும் மனத்தின் விளைவுகளே ஆகும்.	இன்பமும் துன்பமும் மனத்தின் விளைவுகளே. *"Happiness and sorrow are products of the mind."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
12	**Formal Prose**	மனித வாழ்க்கை அறம் சார்ந்த வாழ்க்கை ஆகும்.	மனித வாழ்க்கை அறம் சார்ந்த வாழ்க்கை. *"Human life is a life of righteousness."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**
13	**Formal Prose**	அந்தக் கிராமத்தின் பெயர் கீழடி ஆகும்.	அந்தக் கிராமத்தின் பெயர் கீழடி. *"The name of that village is Keezhadi."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகும்**

#	H1: Sub-Register	Overt Copula Form (Tamil)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Verb
14	**Formal Prose**	திருவள்ளுவர் ஒரு தலைசிறந்த புலவர் ஆவார்.	திருவள்ளுவர் ஒரு தலைசிறந்த புலவர். *"Thiruvalluvar is an eminent poet." — Copula agrees with an honorific subject*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* ***ஆவார்***
15	**Formal Prose**	நான் இந்நாட்டின் குடிமகன் ஆவேன்.	நான் இந்நாட்டின் குடிமகன். *"I am a citizen of this country." — First-person agreement on the copula*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* ***ஆவேன்***
16	**Formal Prose**	இவர்கள் மரியாதைக்கு உரியவர்கள் ஆவர்.	இவர்கள் மரியாதைக்கு உரியவர்கள். *"These people are worthy of respect." — Zero form uses a person-marked noun (உரியவர்கள்)*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* ***ஆவர்***
17	**Formal Prose**	காலத்தின் பெரும் பிரிவுகள் யுகம், கற்பம், நூற்றாண்டு ஆகியவை ஆகும்.	காலத்தின் பெரும் பிரிவுகள் யுகம், கற்பம், நூற்றாண்டு. *"The major divisions of time are ages, aeons and centuries."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* ***ஆகும்***
18	**Formal Prose**	இடைநிலைப் பிரிவுகள் பருவம், ஆண்டு, மாதம் ஆகியவை ஆகும்.	இடைநிலைப் பிரிவுகள் பருவம், ஆண்டு, மாதம். *"The intermediate divisions are seasons, years and months."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* ***ஆகும்***
19	**Formal Prose**	மரியாதை என்பது பண்பு, பதவி, மூப்பு ஆகியவற்றின் அடிப்படையில் பெறப்படும் மதிப்பு ஆகும்.	மரியாதை என்பது பண்பு, பதவி, மூப்பு ஆகியவற்றின் அடிப்படையில் பெறப்படும் மதிப்பு. *"Respect means the regard received on the basis of character, rank and seniority." — Standard formal definition frame: X என்பது Y ஆகும்*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* ***ஆகும்***
20	**Formal Prose**	இத்தகைய மனிதர்களின் இயல்பு மிகவும் மேலோட்டமாக உள்ளது.	இத்தகைய மனிதர்களின் இயல்பு மிகவும் மேலோட்டமானது. *"The nature of such people is very shallow." — Zero form needs the nominalised adjective -ஆனது*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* ***உள்ளது***
21	**Formal Prose**	மரியாதை அனைவருக்கும் அவசியமாக உள்ளது.	மரியாதை அனைவருக்கும் அவசியம். *"Respect is necessary for everyone."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* ***உள்ளது***
22	**Formal Prose**	இவ்வூரின் கோயில் மிகவும் பழமையானதாக உள்ளது.	இவ்வூரின் கோயில் மிகவும் பழமையானது. *"The temple of this town is very old."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* ***உள்ளது***
23	**Formal Prose**	இயற்கைக் காட்சிகள் மிகவும் அழகாக உள்ளன.	இயற்கைக் காட்சிகள் மிகவும் அழகானவை. *"The natural scenes are very beautiful." — Plural agreement: -ஆனவை*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* ***உள்ளன***

#	H1: Sub-Register	Overt Copula Form (Tamil)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Verb
24	**Formal Prose**	அவளுடைய தந்தை மிகவும் கண்டிப்பானவராக இருக்கிறார்.	அவளுடைய தந்தை மிகவும் கண்டிப்பானவர். *"Her father is very strict." — Zero type attested in Kalki, Poiman Karadu*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* ***இருக்கிறார்***
25	**Formal Prose**	காலத்தின் இப்பிரிவுகள் தனித்தனிச் சிறப்பியல்புகளைக் கொண்டவையாக உள்ளன.	காலத்தின் இப்பிரிவுகள் தனித்தனிச் சிறப்பியல்புகளைக் கொண்டவை. *"These divisions of time have particular characteristics."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* ***உள்ளன***
26	**Formal Prose**	மூடநம்பிக்கை உலகின் மிகப்பெரிய சுமையாக உள்ளது.	மூடநம்பிக்கை உலகின் மிகப்பெரிய சுமை. *"Superstition is the heaviest burden on the world."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* ***உள்ளது***
27	**Formal Prose**	இன்றைய தமிழகத்திற்கு இது ஒரு தீவிரமான சமூகப் பிரச்சினையாக உள்ளது.	இன்றைய தமிழகத்திற்கு இது ஒரு தீவிரமான சமூகப் பிரச்சினை. *"It is a serious social problem for present-day Tamil Nadu."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* ***உள்ளது***
28	**Formal Prose**	உலகில் பலவகையான மனிதர்கள் உள்ளனர்.	—— *"There are many kinds of people in the world." — Existential copula cannot be dropped*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உள்ளனர்***
29	**Formal Prose**	ஒவ்வொரு காலப்பிரிவின் தொடக்கத்திலும் நினைவுகூரத்தக்க ஒரு நிகழ்வு உள்ளது.	—— *"There is a memorable event at the start of each division of time."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உள்ளது***
30	**Formal Prose**	அந்தக் கிராமத்தின் மறுமலர்ச்சிக்குப் பின்னால் ஒருவரின் அயராத உழைப்பு உள்ளது.	—— *"Behind the revival of that village lies one person's tireless effort."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உள்ளது***
31	**Formal Prose**	நம்மிடையே இன்னொரு போக்கும் உண்டு.	—— *"There is another tendency among us."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உண்டு***
32	**Formal Prose**	எல்லாவற்றிற்கும் ஒரு தொடக்கம் உண்டு.	—— *"Everything has a beginning."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உண்டு***
33	**Formal Prose**	எல்லாவற்றிற்கும் ஒரு முடிவு உண்டு.	—— *"Everything has an end."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உண்டு***
34	**Formal Prose**	உலகில் மதிக்கத்தக்க மனிதர்கள் பலர் உள்ளனர்.	—— *"There are many respected people in the world."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உள்ளனர்***
35	**Formal Prose**	நம் தனிப்பட்ட வாழ்விலும் சமூக வாழ்விலும் மரியாதைக்கு இன்றியமையாத இடம் உண்டு.	—— *"Respect has an essential place in both our personal and social lives."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உண்டு***

#	H1: Sub-Register	Overt Copula Form (Tamil)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Verb
36	**Formal Prose**	மனித மனத்தில் இன்பமும் துன்பமும் உள்ளன.	—— *"Happiness and sorrow exist in the human mind."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உள்ளன***
37	**Formal Prose**	நூலகத்தில் மூன்று அலமாரிகள் நிறைய நூல்கள் உள்ளன.	—— *"In the library there are three cupboards full of books."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***உள்ளன***
38	**Formal Prose**	இந்நாட்டிற்காகவே நான் உயிரோடு இருக்கிறேன்.	—— *"I am alive for this country alone."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* ***இருக்கிறேன்***
39	**Formal Prose**	அவருக்கு ஒரு மகள் இருக்கிறாள்.	அவருக்கு ஒரு மகள். *"He has a daughter." — Zero possessive possible in narration*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* ***இருக்கிறாள்***
40	**Formal Prose**	அவர்களுக்கு ஒரே குதூகலமாக இருக்கிறது.	அவர்களுக்கு ஒரே குதூகலம். *"They are overjoyed." — Zero type attested in Akilan, Sagotharar Anro*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* ***இருக்கிறது***
41	**Formal Prose**	உண்மையில் பில்லி சூனியம் என்று எதுவும் இல்லை.	—— *"In reality, there is no such thing as witchcraft." — Negative copula cannot be dropped*	Locative/Existential	Negative	Retained (Overt = 0) *Verb:* ***இல்லை***
42	**Formal Prose**	காலத்தின் அளவு கடிகாரத்தின் முட்களில் இல்லை.	—— *"The measure of time is not in the hands of a clock."*	Locative/Existential	Negative	Retained (Overt = 0) *Verb:* ***இல்லை***
43	**Formal Prose**	மனிதர்களின் வாழ்வில் நிம்மதி இல்லை.	—— *"There is no peace in people's lives."*	Locative/Existential	Negative	Retained (Overt = 0) *Verb:* ***இல்லை***
44	**Formal Prose**	மனிதர்களின் உயிருக்கும் உடைமைக்கும் பாதுகாப்பு இல்லை.	—— *"People's lives and property are not safe."*	Locative/Existential	Negative	Retained (Overt = 0) *Verb:* ***இல்லை***
45	**Formal Prose**	இது வியப்புக்குரிய செய்தி அல்ல.	—— *"This is not a matter of wonder."*	Equational/Nominal	Negative	Retained (Overt = 0) *Verb:* ***அல்ல***
46	**Formal Prose**	அவை மலர்கள் அல்ல.	—— *"Those are not flowers."*	Equational/Nominal	Negative	Retained (Overt = 0) *Verb:* ***அல்ல***
47	**Formal Prose**	அவர் அவ்வளவு முதியவர் அல்லர்.	—— *"He is not that old." — Person-agreeing negative copula*	Equational/Nominal	Negative	Retained (Overt = 0) *Verb:* ***அல்லர்***
48	**Formal Prose**	மரியாதைக் குறைவு சமூகத்திற்கு நல்லதல்ல.	—— *"Lack of respect is not good for society." — அல்ல fused with நல்லது*	Adjectival	Negative	Retained (Overt = 0) *Verb:* ***அல்ல***
49	**Formal Prose**	இது வியப்புக்குரிய செய்தி அல்லவா?	—— *"Isn't this a matter of wonder?" — Tag built on the negative copula; cannot be dropped*	Equational/Nominal	Negative	Retained (Overt = 0) *Verb:* ***அல்லவா***
50	**Formal Prose**	காலம் விலைமதிப்பற்ற செல்வம் அல்லவா?	—— *"Isn't time invaluable wealth?"*	Equational/Nominal	Negative	Retained (Overt = 0) *Verb:* ***அல்லவா***
51	**Formal Prose**	பணிவு ஒரு மனப்பண்பு ஆகுமா?	பணிவு ஒரு மனப்பண்பா? *"Is humility a quality of the mind?" — Question clitic -ஆ moves onto the noun*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* ***ஆகுமா***

#	H1: Sub-Register	Overt Copula Form (Tamil)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Verb
52	**Formal Prose**	மூடநம்பிக்கை ஒரு சமூக நோய் ஆகுமா?	மூடநம்பிக்கை ஒரு சமூக நோயா? *"Is superstition a social disease?"*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **ஆகுமா**
53	**Formal Prose**	மரியாதை அனைவருக்கும் அவசியமாக உள்ளதா?	மரியாதை அனைவருக்கும் அவசியமா? *"Is respect necessary for everyone?"*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **உள்ளதா**
54	**Informal Dialogue**	——	அவ என் தங்கச்சி. *"She is my younger sister." — Spoken Tamil has no overt copula for identity*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:*
55	**Informal Dialogue**	——	இது என் புக்கு. *"This is my book."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:*
56	**Informal Dialogue**	——	அது எங்க பழைய காரு. *"That is our old car."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:*
57	**Informal Dialogue**	——	இது உன்னோட சொந்த முடிவு. *"This is your own decision."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:*
58	**Informal Dialogue**	——	அவரு எங்க பக்கத்து வீட்டுக்காரரு. *"He is our neighbour."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:*
59	**Informal Dialogue**	இது ஒரு பெரிய பிரச்சனையா இருக்கு.	இது ஒரு பெரிய பிரச்சனை. *"This is a big problem." — Overt form natural only for evaluations, roles and states*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
60	**Informal Dialogue**	இது எனக்கு ஒரு நல்ல சான்ஸா இருக்கு.	இது எனக்கு ஒரு நல்ல சான்ஸ். *"This is a good opportunity for me."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
61	**Informal Dialogue**	அவரு எங்க ஊரு டாக்டரா இருக்காரு.	அவரு எங்க ஊரு டாக்டரு. *"He is our village's doctor." — Overt -ஆ இருக்காரு = 'works as'*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்காரு**
62	**Informal Dialogue**	ப்ரியா எங்க டீமுக்கு லீடரா இருக்கா.	ப்ரியா எங்க டீம் லீடர். *"Priya is the leader of our team."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கா**
63	**Informal Dialogue**	அவ எங்க கிளாஸுக்கு மானிட்டரா இருக்கா.	அவ எங்க கிளாஸ் மானிட்டர். *"She is the monitor of our class."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கா**
64	**Informal Dialogue**	நான் தான் இந்த வேலைக்கு பொறுப்பா இருக்கேன்.	நான் தான் இந்த வேலைக்கு பொறுப்பு. *"I am the one responsible for this work."*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கேன்**
65	**Informal Dialogue**	வெயில் ரொம்ப அதிகமா இருக்கு.	வெயில் ரொம்ப அதிகம். *"It is very hot (the sun is very strong)."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
66	**Informal Dialogue**	இந்த வேலை ஈஸியா இருக்கு.	இந்த வேலை ஈஸி. *"This work is easy."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**

#	H1: Sub-Register	Overt Copula Form (Tamil)	Zero-Copula Form & English Translation	H2: Predicate Type	H3: Polarity	Assigned Copula Status & Verb
67	**Informal Dialogue**	அந்த மனுஷன் ரொம்ப பாசமா இருக்காரு.	அந்த மனுஷன் ரொம்ப பாசமானவரு. *"That man is very affectionate." — Zero form needs the person-marked -ஆனவரு*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்காரு**
68	**Informal Dialogue**	இந்த இடம் ரொம்ப அழகா இருக்கு.	இந்த இடம் ரொம்ப அழகு. *"This place is very beautiful."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
69	**Informal Dialogue**	அவ ரொம்ப சாதுவா இருக்கா.	அவ ரொம்ப சாது. *"She is very calm."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கா**
70	**Informal Dialogue**	இந்தப் படம் ரொம்ப நல்லா இருக்கு.	இந்தப் படம் சூப்பர். *"This film is very good." — நல்லா is adverbial, so the zero form needs a nominal predicate*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
71	**Informal Dialogue**	இன்னைக்கு வானம் முழுக்க மேகமூட்டமா இருக்கு.	இன்னைக்கு வானம் முழுக்க மேகமூட்டம். *"The sky is completely cloudy today."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
72	**Informal Dialogue**	நீ சொல்றது கொஞ்சம் விசித்திரமா இருக்கு.	நீ சொல்றது கொஞ்சம் விசித்திரம். *"What you are saying is a little strange."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
73	**Informal Dialogue**	இந்த டிரஸ் ரொம்ப சாஃப்டா இருக்கு.	இந்த டிரஸ் ரொம்ப சாஃப்ட். *"This dress is very soft."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
74	**Informal Dialogue**	ரோடு இன்னைக்கு வழுக்கலா இருக்கு.	ரோடு இன்னைக்கு ஒரே வழுக்கல். *"The road is slippery today."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
75	**Informal Dialogue**	அவன் ரொம்ப டீசன்ட்டா இருக்கான்.	அவன் ரொம்ப டீசன்ட். *"He is very decent."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கான்**
76	**Informal Dialogue**	இந்த ஸ்வீட் ரொம்ப இனிப்பா இருக்கு.	இந்த ஸ்வீட் ரொம்ப இனிப்பு. *"This sweet is very sweet."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
77	**Informal Dialogue**	அவ குரல் கொஞ்சம் ஹஸ்கியா இருக்கு.	அவ குரல் கொஞ்சம் ஹஸ்கி. *"Her voice is a little husky."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
78	**Informal Dialogue**	இந்தப் பிரச்சனை ரொம்ப அவசரமா இருக்கு.	இந்தப் பிரச்சனை ரொம்ப அவசரம். *"This problem is very urgent."*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
79	**Informal Dialogue**	எனக்கு ரொம்ப பயமா இருக்கு.	எனக்கு ரொம்ப பயம். *"I am very scared." — Overt form attested in Coolie*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
80	**Informal Dialogue**	நான் வீட்ல இருக்கேன்.	நான் வீட்ல. *"I am at home." — Zero locative is natural mainly as a reply*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கேன்**
81	**Informal Dialogue**	அம்மா கிச்சன்ல இருக்காங்க.	அம்மா கிச்சன்ல. *"Mother is in the kitchen."*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்காங்க**
82	**Informal Dialogue**	அவ இப்போ வீட்ல இருக்கா.	அவ இப்போ வீட்ல. *"She is at home now."*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கா**

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83	**Informal Dialogue**	பசங்க கிளாஸுக்கு வெளியே இருக்காங்க.	பசங்க கிளாஸுக்கு வெளியே. *"The students are outside the classroom."*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்காங்க**
84	**Informal Dialogue**	உன் சாவி என் கையில இருக்கு.	உன் சாவி என் கையில. *"Your key is with me."*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
85	**Informal Dialogue**	அவனோட ரூம் முதல் மாடில இருக்கு.	அவனோட ரூம் முதல் மாடில. *"His room is on the first floor." — Overt type attested in Jagame Thandhiram*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
86	**Informal Dialogue**	வெளியே ஆளுங்க இருக்காங்க.	—— *"There are people outside." — Bare existential cannot be dropped*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* **இருக்காங்க**
87	**Informal Dialogue**	என் கிட்ட ஒரு ஐடியா இருக்கு.	—— *"I have an idea."*	Locative/Existential	Affirmative	Retained (Overt = 0) *Verb:* **இருக்கு**
88	**Informal Dialogue**	நாளைக்கு மீட்டிங் இருக்கு.	நாளைக்கு மீட்டிங். *"There is a meeting tomorrow." — Zero type attested in Coolie*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
89	**Informal Dialogue**	உங்களுக்கு ஒரு குட் நியூஸ் இருக்கு.	உங்களுக்கு ஒரு குட் நியூஸ்! *"There is good news for you."*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்கு**
90	**Informal Dialogue**	அவருக்கு ரெண்டு பொண்ணுங்க இருக்காங்க.	அவருக்கு ரெண்டு பொண்ணுங்க. *"He has two daughters."*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* **இருக்காங்க**
91	**Informal Dialogue**	அவ டாக்டர் இல்ல.	—— *"She is not a doctor." — Spoken இல்ல covers formal அல்ல; cannot be dropped*	Equational/Nominal	Negative	Retained (Overt = 0) *Verb:* **இல்ல**
92	**Informal Dialogue**	அவ எங்க கிளாஸ் மானிட்டர் இல்ல.	—— *"She is not the monitor of our class."*	Equational/Nominal	Negative	Retained (Overt = 0) *Verb:* **இல்ல**
93	**Informal Dialogue**	அந்த விஷயம் உண்மை இல்ல.	—— *"That statement is not true."*	Adjectival	Negative	Retained (Overt = 0) *Verb:* **இல்ல**
94	**Informal Dialogue**	இன்னைக்கு வெதர் நல்லா இல்ல.	—— *"The weather is not good today."*	Adjectival	Negative	Retained (Overt = 0) *Verb:* **இல்ல**
95	**Informal Dialogue**	இது ஒண்ணும் ஈஸி இல்ல.	—— *"This is not easy at all."*	Adjectival	Negative	Retained (Overt = 0) *Verb:* **இல்ல**
96	**Informal Dialogue**	அவரு இன்னைக்கு ஆபீஸ்ல இல்ல.	—— *"He is not at the office today."*	Locative/Existential	Negative	Retained (Overt = 0) *Verb:* **இல்ல**
97	**Informal Dialogue**	என் கிட்ட இப்போ கேஷ் இல்ல.	—— *"I don't have cash with me now."*	Locative/Existential	Negative	Retained (Overt = 0) *Verb:* **இல்ல**
98	**Informal Dialogue**	இங்க எந்தப் பிரச்சனையும் இல்ல.	—— *"There is no problem here."*	Locative/Existential	Negative	Retained (Overt = 0) *Verb:* **இல்ல**
99	**Informal Dialogue**	——	இது உன் ஃபோனா? *"Is this your phone?" — Question clitic -ஆ attaches to the noun; no overt copula*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:*
100	**Informal Dialogue**	——	இவன் தான் உன் புது ஃப்ரெண்டா? *"Is this your new friend?"*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:*

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101	**Informal Dialogue**	——	அவரு உங்க புது மேனேஜரா? *"Is he your new manager?"*	Equational/Nominal	Affirmative	Deleted (Zero = 1) *Verb:*
102	**Informal Dialogue**	அவ ரொம்ப அழகா இருக்காளா?	அவ ரொம்ப அழகா? *"Is she very beautiful?"*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* ***இருக்காளா***
103	**Informal Dialogue**	இன்னைக்கு வெயில் அதிகமா இருக்கா?	இன்னைக்கு வெயில் அதிகமா? *"Is it very hot today?"*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* ***இருக்கா***
104	**Informal Dialogue**	ரோடு இன்னைக்கு காலியா இருக்கா?	ரோடு இன்னைக்கு காலியா? *"Is the road empty today?"*	Adjectival	Affirmative	Deleted (Zero = 1) *Verb:* ***இருக்கா***
105	**Informal Dialogue**	உன் ஃபோன் இப்போ ஆன்ல இருக்கா?	உன் ஃபோன் இப்போ ஆன்லயா? *"Is your phone switched on now?"*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* ***இருக்கா***
106	**Informal Dialogue**	அவரு இப்போ வீட்ல இருக்காரா?	அவரு இப்போ வீட்லயா? *"Is he at home now?"*	Locative/Existential	Affirmative	Deleted (Zero = 1) *Verb:* ***இருக்காரா***